The Distributional Dividend of Macroprudential Policy

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Abstract

Prudential regulation is designed for financial stability, yet the credit cycles it leans against affect households unevenly. We trace the distributional footprint of these regulations in a panel of countries over 1999–2024 and find that tighter policies lower income inequality. The effects are gradual and clearer for instruments aimed at lenders, and are stronger on inequality measured after redistribution. To rationalize this, we build a model with financial frictions and capital-skill complementarity in which prudential policy taxes the returns to intermediation. A prudential tightening restrains capital accumulation and compresses the skilled-unskilled wage gap, reducing inequality when the wage gap falls proportionally more than aggregate income, a condition satisfied in our calibration. Across alternative policy rules, financial stabilization schemes with countercyclical features can moderate inequality as a by-product, whereas explicitly targeting inequality need not improve distributional outcomes. Financial stability and equality can be complementary, but their alignment depends on the design of prudential regulation.

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Key words: Macroprudential policy, Income inequality, Financial frictions, Capital-skill complementarity, Banking intermediation

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1 Introduction

Macroprudential regulation has become a standard component of the policy toolkit since the global financial crisis (GFC), as frameworks such as Basel III and the IMF's Integrated Policy Framework have consolidated a set of instruments that are deployed routinely to contain financial vulnerabilities. Their effectiveness on the intended targets has been evaluated extensively, along with their macroeconomic costs and cross-border leakages.[1] Their consequences for the distribution of income, by contrast, remain comparatively under-explored, and the evidence that exists is sparse: depending on the sample, the instruments, and the prevailing economic conditions, prudential measures have been linked to higher, lower, or state-contingent inequality, and only rarely to an explicit account of why regulation and inequality should comove at all. It seems relevant, then, to determine whether macroprudential regulations systematically shape the income distribution, and—most importantly—through which mechanisms.

We explore these questions in this paper, and organize the analysis around three of them. Does a tightening of macroprudential regulation raise or lower income inequality? Which segments of the prudential toolkit drive the relationship? And does it survive the fiscal redistribution that separates market from disposable income? Crucially, unveiling a mechanism behind these responses is central to our analysis, as it allows us to gauge whether a distributional effect is a by-product of prudential policy or a lever that regulators could, and should, in principle, account for deliberately. These questions have gained relevance as inequality has moved to the center of policy conversations, while remaining largely unaccounted for by financial regulators, whose mandate is primarily concerned with financial stability. There is, moreover, a latent tension between the intermediation that fuels growth and the stability that regulation secures; whether restraining the former also carries a distributional dividend is, we argue, an open and consequential question.

We proceed on two fronts, empirical and theoretical. On the empirical side, we estimate lag-augmented local projections (Jordà, 2005) on an annual panel of advanced, emerging, and developing economies over 1999–2024, combining the IMF's Integrated Macroprudential Policy database (iMaPP; Alam, Alter, Eiseman, Gelos, Kang, Narita, Nier, and Wang, 2025) with market and disposable-income Gini coefficients from the Standardized World Income Inequality Database (Solt, 2020). Local projections suit this setting: they are robust to dynamic misspecification (Montiel Olea and Plagborg-Møller, 2021) and, more importantly, recover the

[1] See Richter, Schularick, and Shim (2019); Alam, Alter, Eiseman, Gelos, Kang, Narita, Nier, and Wang (2025); Chari, Dilts-Stedman, and Forbes (2022), among others.

long-horizon dynamics that slow-moving inequality measures display. Beyond an aggregate net-tightening index, we exploit the granularity of iMaPP to separate tightenings from loosenings and to disaggregate the toolkit by the type of risk addressed, by policy objective, and by the agent targeted (Claessens et al., 2013; Cabral et al., 2019; BIS, 2012), which locates the effect within the toolkit rather than attributing it to macroprudential policy in the aggregate.

Our empirical results indicate a negative relationship. A macroprudential tightening tends to lower income inequality, with the effect building gradually over the horizon rather than on impact. It is present for both the market and the disposable Gini, and stronger and more precisely estimated for the latter, consistent with fiscal redistribution reinforcing—rather than offsetting—the inequality-reducing channel of prudential policy, in line with Guerello (2018) for unconventional monetary policy. The response is driven by active tightenings and, across instrument classifications, by lender-targeted, capital-based, resilience, and cycle-management tools—those that most directly constrain the scale, leverage, and profitability of banking intermediation—whereas borrower-targeted caps on loan-to-value and debt-service-to-income ratios are close to zero and less precisely estimated. The pattern holds across advanced, emerging, and developing economies, materializing earlier in the former, and survives restricting the sample to the post-crisis period.

To rationalize this evidence, we develop a closed-economy general equilibrium model with two types of agents – capital-owning entrepreneurs who supply skilled labor, and households who supply unskilled labor and save through banks. A banking sector intermediates funds subject to agency frictions à la Gertler and Karadi (2011), production features capital-skill complementarity in the nested-CES form of Krusell et al. (2000), and macroprudential policy is modeled as a tax on the returns to banking intermediation—a general enough setup to stand in for a range of prudential instruments. The environment is in the spirit of Liu, Spiegel, and Zhang (2023a), who study capital-flow regulation, and of Dolado, Motyovszki, and Pappa (2021), whose capital-skill-complementarity production side we adapt. We turn this framework toward domestic prudential regulation and characterize its distributional implications analytically. The model admits a transparent inequality statistic: at the steady state the two-agent Gini reduces to the gap between the entrepreneurs' income share (Λ_e) and their population share ($1 - \mu$), so that any intervention that compresses the entrepreneurial income share lowers measured inequality one-for-one.

Within this framework, a permanent tightening of the policy instrument reduces steady-state inequality locally under a transparent sufficient condition. A tightening contracts the steady-

state capital stock and, through capital–skill complementarity, compresses the skilled–unskilled wage gap. Inequality falls when this compression in the wage gap is sufficiently strong relative to the change in aggregate disposable income. This mechanism ties the distributional incidence of prudential policy to the same capital–skill complementarity that renders changes in capital and financial intermediation skill-biased in Liu et al. (2023a) and Dolado et al. (2021). We show that the sufficient condition is satisfied at the baseline calibration, where the wage-gap response dominates by a substantial margin. The result provides a structural rationale for the inequality-reducing effects of macroprudential tightening observed empirically and in the quantitative analysis.

Comparing policy schemes that respond to lending, output, inequality, or bank leverage (Angelini et al., 2014; Hecker et al., 2024), we further find that policies leaning against the business cycle yield a distributional dividend that exceeds that of setups targeting more complex financial quantities with weaker countercyclical properties, or implementing equality explicitly as a secondary target. In the case of a leverage-based rule, for example, a tightening policy may turn into a de facto subsidization scheme and thereby raise inequality. On the other hand, explicit inequality objectives may weaken or even offset the inequality-reducing effects of policies that temper financial intermediation, particularly when stabilizing inequality calls for an instrument adjustment that runs counter to the response warranted by financial conditions. In such cases, incorporating inequality directly into the policy rule can compromise its effectiveness in stabilizing both financial conditions and the income distribution. Taken together, these results point to the moderation of inequality as a by-product of prudential policy under standard rules that smooth the credit cycle: in curbing an intermediation boom, regulators can also temper the distributional consequences of heightened banking activity, and thus financial-stability and inequality objectives need not be in conflict.

Related literature. Our paper connects to several strands of work. The first is the empirical literature on macroprudential policy and inequality, which has not settled on a sign. Frost and van Stralen (2018) report a positive association between the use of several prudential instruments and income inequality; their measures, however, predate iMaPP—largely use dummies, without a tightening-loosening distinction—and their estimates are static, and thus are limited in capturing inequality dynamics, given the slow-moving nature of Gini measures. Georgescu and Vila Martin (2021) and Carpantier et al. (2018) narrow the lens to borrower-based tools and reach state-contingent conclusions, in which caps on loan-to-value and debt-service-to-income ratios can raise inequality at introduction yet dampen its rise under

adverse conditions. We differ on the sign of the aggregate relationship, on the breadth and directionality of the measures—our updated iMaPP sample shows the effect is driven by the lender-targeted instruments these studies largely abstract from—and in recovering dynamic responses through local projections.

A second strand supplies the structural mechanisms our theory builds on. Korinek and Kreamer (2014) place distributive conflict at the center of financial regulation, showing that deregulation allows the financial sector to take on more risk than is socially optimal, raising its surplus share at the expense of the rest of the economy—an inequality-increasing force countered by prudential oversight. Rubio and Unsal (2020) find that a well-calibrated active prudential rule is equalizing whereas overly conservative policies in the presence of noisy information are not. The distributional bite of our mechanism also runs through capital-skill complementarity, following Krusell, Ohanian, Ríos-Rull, and Violante (2000), Dolado et al. (2021), and Liu et al. (2023a), who establish that capital deepening is skill-biased in monetary and capital-flow settings; we carry the same channel into a prudential one, where it is the scale of banking intermediation that moves the skill premium. A related literature runs the arrow in reverse, from inequality to financial fragility and bank risk (Mitkov, 2020; Mitkov and Schüwer, 2025), implying that the distribution is jointly determined with stability, consistent with the complementarity our results suggest. Relative to this strand, our contribution is to make that mechanism explicit and tractable—a two-agent economy in which the macroprudential tax compresses both financial rents and the skill premium—uniting the distributive-conflict view of regulation with the capital-skill-complementarity channel.

Finally, our work also relates to the longer-standing literature on monetary policy and inequality, pertinent because post-crisis unconventional monetary policy—operating through bank balance sheets—can be read as a de facto macroprudential intervention. The evidence there points, on balance, to an equalizing effect of monetary expansions: contractionary shocks raise inequality in the United States (Coibion et al., 2017), expansions reduce it across a large panel and asymmetrically (Furceri et al., 2018), and the euro-area effect is equalizing on average but heterogeneous and redistribution-dependent (Guerello, 2018); even non-standard easing need not raise inequality, as activity gains at the bottom offset opposing financial-income effects (Casiraghi et al., 2018), while Voinea et al. (2018) add the reverse-channel counterpart in which inequality shapes policy transmission. Monnin (2017) stresses how scarce the parallel evidence on macroprudential policy remains. Our results help close that gap, extending the equalizing reading from the monetary to the prudential toolkit and supplying a microfounded mechanism.

The remainder of the paper is organized as follows. Section 2 describes the data and the local-projection strategy and presents the empirical results. Section 3 develops the general equilibrium model and derives the model-implied Gini coefficient and its comparative statics with respect to the prudential instrument. Section 4 discusses the calibration, and Section 5 evaluates the dynamic implications of alternative macroprudential rules. Finally, Section 6 concludes.

2 Empirical strategy and stylized facts

To motivate our research question and the subsequent conceptual analysis, we begin by examining the empirical relationship between macroprudential regulation and income inequality. Using a panel of countries over the period 1999–2024, we estimate local projection impulse responses of inequality measures to macroprudential policy interventions, conditioning on a set of domestic macroeconomic controls. In this section, we describe the data, outline the empirical methodology, and present the main results.

2.1 Data

We construct an annual panel dataset covering a broad set of advanced, emerging, and developing economies. The dataset merges several sources to obtain measures of income inequality, macroprudential policy stances, and macroeconomic controls. The sample period runs from 1999 to 2024, and we retain only country-year observations for which at least ten years of inequality data are available. Countries with no recorded macroprudential interventions during the sample period are excluded.[2]

Inequality measures. Our main dependent variables are two measures of income inequality from the Standardized World Income Inequality Database (SWIID) (Solt, 2020). The first is the Gini coefficient of market income (*Gini market*), which captures pre-tax and pre-transfer inequality. The second is the Gini coefficient of disposable income (*Gini disposable*), which measures inequality after taxes and government transfers. Using both measures allows us to assess whether macroprudential regulation affects inequality before or after the fiscal redistribution process.

[2]Specifically, we exclude Burundi, Saint Kitts and Nevis, Yemen, Zambia, Ethiopia, and Sudan.

Macroprudential policy indicators. We obtain macroprudential policy data from the Integrated Macroprudential Policy Database (iMaPP) of the International Monetary Fund, as reported by Alam, Alter, Eiseman, Gelos, Kang, Narita, Nier, and Wang (2025). The database records tightenings and loosenings across 17 individual policy instruments. For each instrument j, country i, and period t, the indicator is defined as:

$$MaPP_{i,t}^{j} = \begin{cases} +1 & \text{if tightened} \\ 0 & \text{if unchanged} \\ -1 & \text{if loosened} \end{cases}.$$

We aggregate across all 17 instruments to obtain the overall macroprudential policy stance:

$$MaP_{i,t} = \sum_{j=1}^{17} MaPP_{i,t}^{j}.$$

This net tightening indicator can be interpreted as a measure of the overall macroprudential stance, with higher values indicating a stricter regulatory posture (Fernández, Rebucci, and Uribe, 2015; Richter, Schularick, and Shim, 2019).[3] In addition to this aggregate index, we consider separate indicators for policy tightenings only ($MaP_{i,t}^{T}$) and loosenings only ($MaP_{i,t}^{L}$), which allow us to distinguish between the inequality effects of regulatory tightening and easing episodes.

Beyond the aggregate and direction-based indicators, we further disaggregate the macroprudential toolkit along three dimensions. The first classification, based on the type of risk addressed, follows Cabral et al. (2019) (ECB) and BIS (2012), and groups the instruments into: (i) *capital-based* tools (countercyclical capital buffer, conservation buffer, capital requirements, leverage ratio, loan loss provisions, and SIFI-specific rules); (ii) *asset-based* tools (limits on credit growth, loan restrictions, loan-to-value ratios, debt-service-to-income ratios, and tax-based instruments); and (iii) *liquidity and FX-based* tools (limits on foreign currency lending, liquidity requirements, loan-to-deposit ratios, limits on FX positions, and reserve requirements).

A second classification, following Claessens, Ghosh, and Mihet (2013), distinguishes instruments by policy objective: (i) *resilience tools* (countercyclical capital buffer, conservation buffer, capital requirements, leverage ratio, loan loss provisions, and liquidity requirements); (ii) *cycle-management tools* (limits on credit growth, loan restrictions, limits on foreign currency

[3] Figures 14–16 in Appendix A plot the macroprudential policy stance by country.

lending, LTV, DSTI, loan-to-deposit ratios, limits on FX positions, and reserve requirements); and (iii) *tax-type and other instruments* (taxes, SIFI regulations, and other measures such as stress-testing requirements).

Finally, a third classification groups instruments by the type of agent targeted: (i) *borrower-targeted* tools (LTV and DSTI ratios); and (ii) *lender-targeted* tools (the remaining 15 instruments in the toolkit).

Control variables. The domestic control variables included in the baseline specification are as follows. A banking crisis dummy, sourced from the Global Macro Database (Müller et al., 2026), is extended to cover a five-year window following the onset of each crisis episode to capture the persistent effects of financial crises on inequality.[4] Real GDP growth is computed as the log-difference of real GDP, also obtained from the Global Macro Database. CPI inflation and the central bank policy rate are taken from the same source; notably, interest rates are allowed to take on negative values during zero-lower-bound episodes, making shadow-rate adjustments unnecessary.[5]

Capital inflows to GDP are constructed from the IMF International Financial Statistics (IFS) database, with the flows normalized by GDP using exchange rate and nominal GDP data from the Global Macro Database. Domestic credit to the private sector (as a share of GDP) is obtained from the World Development Indicators of the World Bank. A financial development index is sourced from the IMF Financial Development Database, and the nominal exchange rate depreciation (vis-à-vis the US dollar) is computed as the log-difference of the bilateral exchange rate from the Global Macro Database.

Variables that are already stationary or that represent rates—such as inflation, the central bank rate, capital inflows to GDP, and the macroprudential policy index—enter in levels. The banking crisis indicator is a dummy variable. Finally, a small number of countries with no recorded macroprudential interventions throughout the sample are excluded from the analysis. Table 2 in Appendix A summarizes the variables, their definitions, and data sources.

2.2 Empirical methodology

We employ a lag-augmented local projection (LP) approach, building on Jordà (2005). LP methods are widely used in applied macroeconomics because they are generally less sensitive

[4]This treatment follows common practice in the literature; see, for example, Laeven and Valencia (2013).

[5]The source of the central bank rates in the Global Macro Database is the Bank for International Settlements (BIS).

to dynamic misspecification than standard VAR-based impulse response estimators (Haug and Smith, 2012; Montiel Olea and Plagborg-Møller, 2021). They are also well suited to estimating medium- and long-horizon responses in persistent data, an important feature in our setting given the slow-moving nature of inequality measures such as the Gini coefficient.

Our baseline specification adopts a long-difference approach with Cholesky-type timing, which allows for delayed effects of the policy variable on inequality.[6] Specifically, for horizons $h = 1, 2, \ldots, H$ (with $H = 10$), we estimate:

$$\text{Gini}_{i,t+h} - \text{Gini}_{i,t} = \alpha^{(h)} + \beta_1^{(h)} \text{MaP}_{i,t-1} + \boldsymbol{\gamma}^{(h)\prime} \mathbf{X}_{i,t} + FE_i^{(h)} + FE_t^{(h)} + \epsilon_{i,t+h}, \tag{1}$$

where $\text{Gini}_{i,t+h}$ is the Gini coefficient (market or disposable) of country i at horizon h, $\text{MaP}_{i,t-1}$ is the lagged macroprudential policy indicator, $\mathbf{X}_{i,t}$ is a vector that stacks the current value and two lags of the domestic controls (banking crisis dummy, real GDP growth, inflation, central bank rate, capital inflows to GDP, domestic credit growth, financial development growth, and exchange rate depreciation), one additional lag of the policy indicator, and two lags of the change in the Gini coefficient, $FE_i^{(h)}$ and $FE_t^{(h)}$ are country and time fixed effects, respectively, and $\epsilon_{i,t+h}$ is the residual. The policy variable enters with one lag to mitigate potential simultaneity bias, as macroprudential decisions are typically predetermined relative to contemporaneous inequality outcomes. The lag-augmented structure includes additional lags of the controls (two lags), of the policy variable (one additional lag beyond the main regressor), and of the dependent variable in differences (two lags), following the ARDL-style augmentation as in Coman and Lloyd (2022) and Richter et al. (2019). The inclusion of lagged dependent variables in differences preserves the long-difference interpretation of the left-hand side while accounting for serial correlation in the inequality dynamics.[7] Standard errors are computed using the panel-corrected standard errors of Beck and Katz (1995), which account for cross-sectional dependence and heteroskedasticity. The resulting impulse response functions report the cumulative change in the Gini coefficient at each horizon in response to a one-unit increase in the macroprudential policy indicator, along with 68% and 95% confidence intervals.

Our main object of interest is the impulse-response function, characterized by the coefficients $\beta_1^{(h)}$, which trace the dynamic response of the Gini coefficient to a macroprudential

[6] Under this format, the dependent variable is the cumulative change in the Gini coefficient from period t to $t + h$, with the policy variable dated $t - 1$, so that its contemporaneous effect on inequality is excluded and the response at $h = 0$ is zero by construction. See Jordà, Singh, and Taylor (2024) for a discussion of both approaches and evidence that the results are typically analogous.

[7] See Piger and Stockwell (2025) for a discussion of the long-difference approach and its advantages in bias reduction and confidence interval coverage.

intervention over the 10-year horizon. The responses are expressed in Gini points per one-unit net macroprudential intervention. A negative estimate means inequality is lower h years after a tightening relative to the period before the intervention.

2.3 Results

We present the results in a sequence that moves from the most aggregated policy measure to progressively finer disaggregations. Throughout, we report impulse responses for both the market Gini and the disposable Gini. The impulse responses display the cumulative change in the Gini coefficient over a 10-year horizon, with darker shaded areas representing 68% (one-standard-deviation) confidence intervals and lighter areas the 95% confidence intervals.

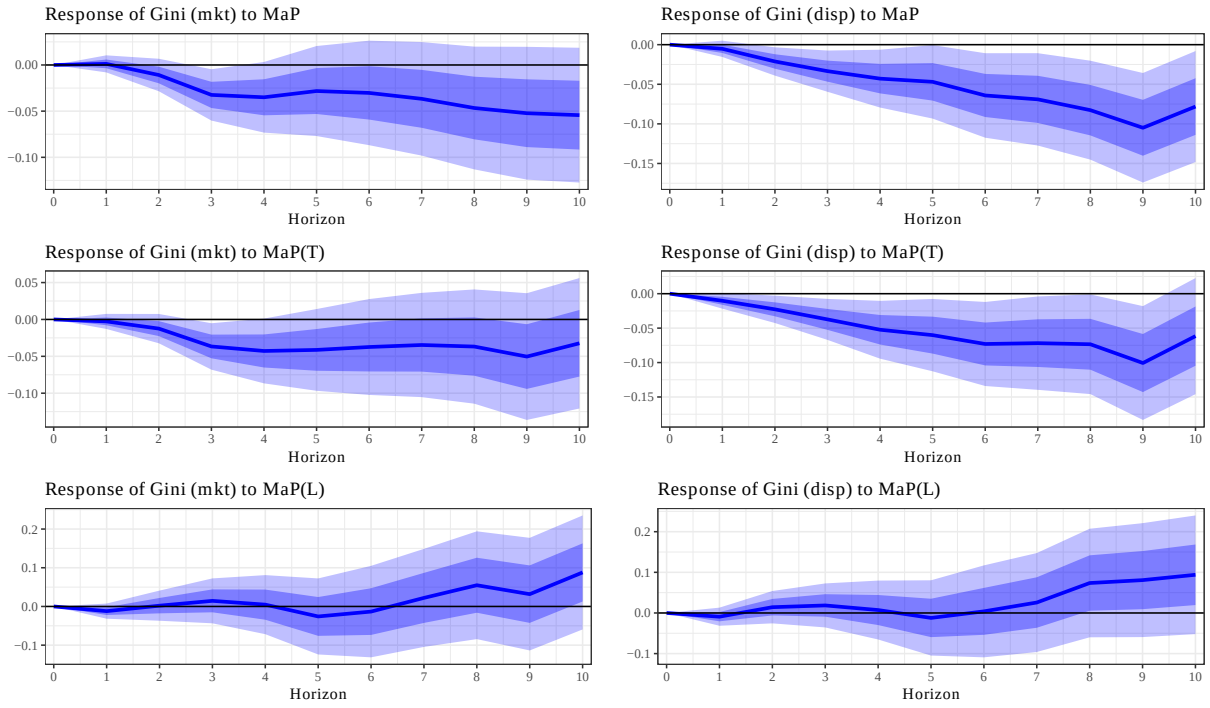
Aggregate macroprudential policy index. Figure 1 reports the impulse response functions of the Gini coefficients to a one-unit increase in the aggregate macroprudential policy index (which considers all 17 instruments and nets tightenings against loosenings). The left column shows the response of the market Gini and the right column that of the disposable Gini.

Both Gini measures respond negatively to a macroprudential tightening, with the effect building gradually over time. For the market Gini (top-left panel), the point estimate declines steadily from zero, reaching approximately -0.06 Gini points around horizon 10. The 68% confidence interval falls below zero from year three onward, and the 95% band becomes significant three years ahead and includes zero thereafter, indicating considerable uncertainty around the estimated response. For the disposable Gini (top-right panel), the response is notably stronger and more precisely estimated: the point estimate declines to approximately -0.10 around horizons 8–9, with the 68% and 95% confidence intervals lying below zero for nearly all horizons. The stronger and more significant response of the disposable Gini may reflect the fact that redistributive policies amplify the inequality-reducing channel of macroprudential regulation, a result consistent with the findings of Guerello (2018) for the case of unconventional monetary policy.

This finding of a negative relationship between macroprudential tightening and inequality is noteworthy. It contrasts with the positive association reported in recent work (e.g., Frost and van Stralen 2018), which finds that stricter regulation is associated with higher inequality for the majority of the prudential policy toolkit, and reinforces the findings of Rubio and Unsal (2020) and Carpantier et al. (2018) suggesting a potential negative feedback between financial regulations and equality. Our result is consistent with a plausible interpretation, also suggested

by Liu et al. (2023a) in the context of capital flows and capital controls: higher-income agents and bank shareholders benefit disproportionately from elevated financial intermediation during normal economic expansions. As the economy grows and financial activity expands, inequality tends to increase because the gains from banking activities accrue asymmetrically to wealthier agents. Macroprudential tightening curtails this process by reducing the scope and profitability of banking intermediation, thereby slowing the mechanism through which financial development widens inequality. In this sense, regulators pursuing financial stability inadvertently moderate the distributional consequences of the credit cycle.

Figure 1: Impulse Responses of the Gini Coefficients to Macroprudential Policy: Aggregate Index, Tightenings, and Loosenings



Notes: Each panel reports the impulse response of the Gini coefficient to a one-unit increase in the respective macroprudential policy index. The solid line is the point estimate; the darker shaded area represents the 68% confidence interval (one standard deviation); the lighter shaded area represents the 95% confidence interval. Standard errors are Beck–Katz panel-corrected standard errors. Top row: aggregate net policy index (MaP). Middle row: tightenings-only index (MaP^T). Bottom row: loosenings-only index (MaP^L). Unlike all the other categories, the loosening index increases (+1) with softenings in the instruments across the macroprudential toolkit (17 instruments). Left column: market Gini. Right column: disposable Gini.

2.3.1 Tightenings versus loosenings

To investigate whether the direction of the policy change matters, we re-estimate the model replacing the aggregate net index with the tightening-only index (MaP^T) and the loosening-only index (MaP^L), separately. The impulse responses are reported in the middle and bottom rows of Figure 1. The tightening-only index (middle row) yields negative impulse responses that closely resemble those from the aggregate index, suggesting that the baseline results are primarily driven by tightening interventions. For the market Gini, the point estimate declines from approximately -0.03 to -0.06 at horizons 3–9, with the 68% confidence band below zero over much of this range. For the disposable Gini, the response is again stronger: the point estimate reaches approximately -0.07 to -0.10 at horizons 6–9, and achieves significance consistently for most horizons.

The loosening-only index, when included in isolation, is transformed into a sum of dummies across instruments by changing the sign of the loosenings entering the aggregate index and zeroing any other interventions. With this format, the loosening index increases with less stringent interventions and thus has the opposite interpretation to the other policy indexes: a positive IRF would indicate a negative relationship between stricter prudential regulations and inequality (an increase in the Gini index). The IRFs for this index are positive, in line with the other results in Figure 1 and in most exercises. However, the responses are not significant even at the 68% confidence level in most horizons. The contrasting significance in this latter case is consistent with the fact that loosening interventions are less frequent and tend to cluster during crisis episodes, which reduces the effective sample for identification. It also suggests that the inequality-reducing effects of macroprudential regulation are driven by active tightenings rather than by a reversal of loosening interventions.

2.3.2 Results by type of instrument: risk-based classification

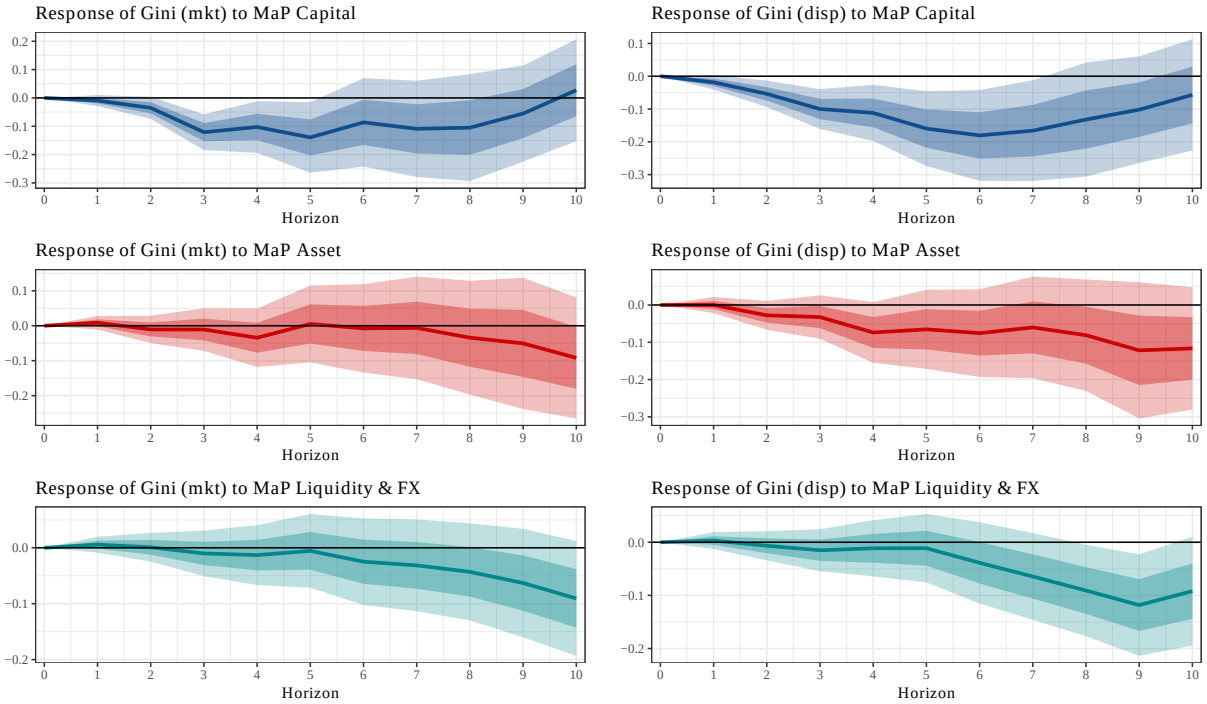
We next disaggregate the macroprudential toolkit by the type of risk targeted, following the classification of Cabral, Detken, Fell, Henry, Hiebert, Kapadia, Pires, Salleo, Constâncio, and Nicoletti Altimari (2019) and BIS (2012). Figure 2 reports the impulse responses for capital-based, asset-based, and liquidity and FX-based instruments, separately for each Gini measure.

Capital-based instruments (top row)—which include countercyclical capital buffers, conservation buffers, general capital requirements, leverage ratios, loan loss provisions, and SIFI-specific rules—produce the largest negative responses among the three categories. For

the market Gini, the point estimate declines sharply to approximately -0.15 at horizons 4–5, before reverting at later horizons. For the disposable Gini, the response is persistently negative, reaching -0.15 to -0.20 at horizons 5–7. Both of these responses are significant. The strength of this response is consistent with the view that capital requirements directly constrain bank leverage and profitability, which is potentially central to the mechanism through which the level of banking intermediation comoves with inequality.

Asset-based instruments (middle row)—comprising limits on credit growth, loan restrictions, LTV caps, DSTI limits, and tax-based tools—yield negative point estimates steadily for both Gini measures. However, the confidence intervals are wider than for capital-based tools, and the 95% bands include zero at most horizons. The effects on disposable Gini are subject to less uncertainty and are significant at one standard deviation for most horizons and close to statistical significance at the two-standard-deviation level for some horizons. These tools operate primarily by restricting the volume and terms of credit extension, which can moderate the accumulation of financial wealth among higher-income borrowers and intermediaries, but their effects seem less precisely identified in the data.

Figure 2: Impulse Responses by Type of Instrument: Risk-Based Classification



Notes: Top row: Capital-based (CCB, Conservation, Capital, LVR, LLP, SIFI). Middle row: Asset-based (LCG, LoanR, LTV, DSTI, Tax). Bottom row: Liquidity and FX (LFC, Liquidity, LTD, LFX, RR). Left column: market Gini. Right column: disposable Gini. The classification follows Cabral et al. (2019) and BIS (2012).

Liquidity and FX-based instruments (bottom row)—including limits on foreign currency lending, liquidity requirements, loan-to-deposit ratios, limits on FX positions, and reserve requirements—produce negative responses for both Gini measures. The market Gini declines to approximately -0.10 at horizon 10, with the 68% confidence band below zero from around horizon 8 onward. The disposable Gini shows a similar pattern, declining for later horizons and achieving significance at horizon 8 and 9. The response is more gradual than that of capital-based instruments but more precisely estimated than asset-based tools. This gradual accumulation is consistent with the broader finding that regulations targeting specific risk exposures—such as capital requirements—tend to produce sharper initial effects than more generalized prudential measures (Chari, Dilts-Stedman, and Forbes, 2022).

2.3.3 Results by policy objective

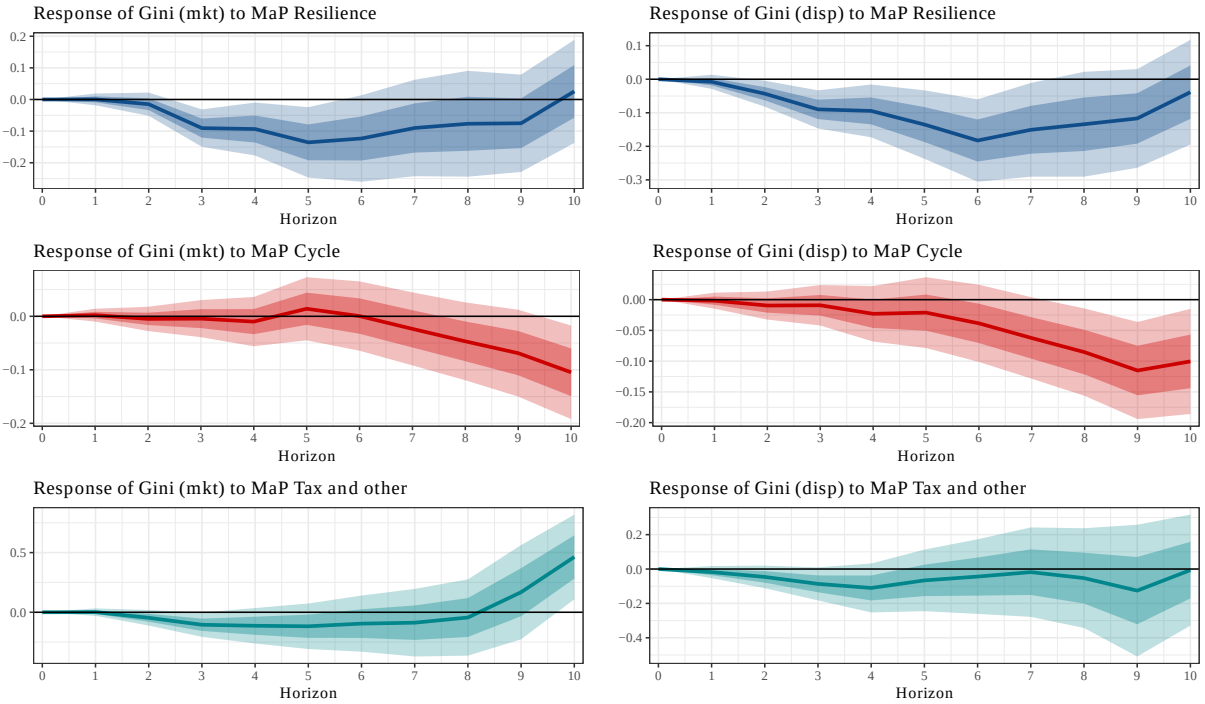
A complementary classification, following Claessens, Ghosh, and Mihet (2013), groups instruments by the structural and cyclical features of the financial system they are designed to address. Figure 3 reports the impulse responses for resilience tools, cycle-management tools, and tax-type and other instruments.

Resilience tools (top row)—those designed to build structural buffers that can be drawn down in times of stress—produce a negative response for both Gini measures that materializes relatively fast. For the market Gini, the point estimate declines to approximately -0.10 to -0.15 at horizons 3–5, with the confidence bands below zero over this range, before reverting toward zero at later horizons. For the disposable Gini, the response is stronger and more persistent, reaching approximately -0.15 to -0.18 at horizons 5–7 with the bands clearly below zero for most of the horizons. The relatively rapid onset of this effect is consistent with the nature of resilience instruments—capital buffers, leverage limits, and provisioning rules—which directly constrain intermediation capacity upon implementation.

Cycle-management tools (middle row)—which aim to lean against the financial cycle by restricting credit growth, tightening lending standards, or adjusting reserve requirements—yield responses that are initially muted but grow steadily more negative over time. For the market Gini, the point estimate remains near zero through horizon 6, then declines to approximately -0.10 by the last horizon, where the 95% confidence interval excludes zero. For the disposable Gini, the decline is more precisely estimated, reaching approximately -0.10 by horizon 10, with the confidence intervals below zero roughly from horizon 7 onward. Although the cycle-management tools ultimately reach a comparable magnitude to the resilience instruments, their transmission is notably more gradual. This pattern is consistent

with the findings of Gambacorta and Murcia (2020), who show—using credit registry data across several economies—that cycle-dampening instruments such as reserve requirements propagate to credit conditions more slowly than resilience-oriented capital measures, the latter taking effect more rapidly given their direct impact on bank balance sheets.

Figure 3: Impulse Responses by Policy Objective



Notes: Top row: Resilience (CCB, Conservation, Capital, LVR, LLP, Liquidity). Middle row: Cycle management (LCG, LoanR, LFC, LTV, DSTI, LTD, LFX, RR). Bottom row: Tax-type and other (Tax, OT, SIFI). Left column: market Gini. Right column: disposable Gini. See the notes to Figure 1 for additional details on the specification. The classification follows Claessens et al. (2013).

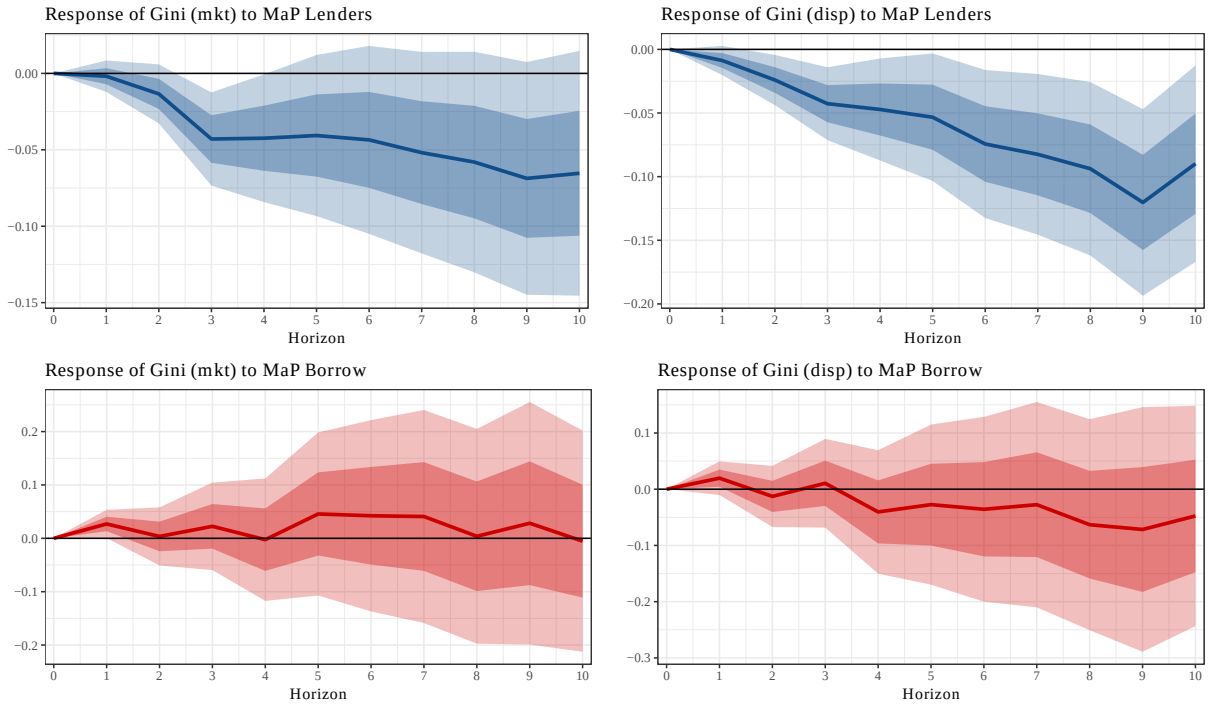
Tax-type and other instruments (bottom row)—including financial sector taxes, SIFI-specific regulations, and other tools—produce impulse responses that are negative on average but estimated with substantial uncertainty. For the disposable Gini, the point estimates reach approximately -0.10 to -0.12 at horizons 4 and 9, but the 95% confidence bands include zero throughout. For the market Gini, the point estimates are slightly negative in the short run and turn positive at later horizons, again with wide confidence intervals. This imprecision is consistent with the relatively low frequency of tax-type prudential interventions in the sample and the heterogeneity of the "other" category, which groups together instruments with potentially diverse transmission mechanisms.[8]

[8]Given how contrasting this positive effect in the final period seems, even if partly related to residual and

2.3.4 Results by targeted agent

Finally, we disaggregate the toolkit by the type of agent targeted by the regulation: lender-targeted instruments (15 tools) versus borrower-targeted instruments (LTV and DSTI ratios). The impulse responses are reported in Figure 4.

Figure 4: Impulse Responses by Targeted Agent



Notes: Top row: Lender-targeted (CCB, Conservation, Capital, LVR, LLP, LCG, LoanR, LFC, Tax, Liquidity, LTD, LFX, RR, SIFI, OT). Bottom row: Borrower-targeted (LTV, DSTI). Left column: market Gini. Right column: disposable Gini.

Lender-targeted instruments (top row) produce a negative response. For the market Gini, the point estimate declines steadily from zero to approximately -0.05 to -0.07. For the disposable Gini, the effect is similarly negative, reaching a significant -0.07 to -0.12 at horizons 6–9. These results are expected, as lender-targeted tools directly constrain bank behavior—capital buffers, leverage limits, credit growth caps, reserve requirements—and thereby reduce the rents accruing to financial intermediaries and their stakeholders. It should be noted that although this result underscores the importance of banking intermediaries in shaping the

uncategorized policies, we inspected its robustness and found that it disappears if we remove emerging economies from the sample, or if we focus on the post-2008 sample for all countries. At the same time, for this method the responses of earlier horizons, typically below the order of the local projection, are deemed more reliable than effect reversals at the very last period (Baumeister, 2025).

inequality effects of regulation, the response can also be interpreted as a filtered version of the aggregate policy stance, as the lender-targeted indicator still encompasses most of the instruments in the toolkit.

Borrower-targeted instruments (bottom row), by contrast, yield responses that are close to zero and statistically insignificant for both Gini measures. For the market Gini, the point estimates are slightly positive at most horizons, while for the disposable Gini they are close to zero in the short run and mildly negative at longer horizons. In both cases, the confidence intervals are wide and include zero throughout. This result is consistent with the notion that borrower-targeted tools such as LTV and DSTI limits restrict credit access for households but do not directly curtail banking profitability or the financial rents that drive the inequality —we will focus on this mechanism in our theoretical framework. The weaker results and the higher uncertainty around these responses may also reflect the smaller number of interventions in this category and the associated small-sample limitations (two instruments only for borrower-targeted tools versus fifteen for lenders).

2.3.5 Robustness exercises

By level of economic development: We re-estimate the baseline specification separately for advanced economies (AEs), emerging market economies (EMs), and developing economies (DEs). The main negative response pattern is preserved across all three groups and instrument categories. The main difference is in timing. The decline in inequality appears earlier in AEs, whereas in EMs and DEs the response materializes at longer horizons. As in the baseline, responses of the disposable Gini are generally more precisely estimated than those of the market Gini, although the latter exhibit more statistically significant horizons in the country-group estimations than in the pooled sample.

Excluding developing economies: Given lower macroprudential interventionism in DEs, we also estimate the model on the sample that includes only AEs and EMs. The estimated impulse responses remain close to the baseline in sign and magnitude.

Post-2008 sample only: Restricting the sample to the post-global-financial-crisis period yields the same qualitative result—a negative response of inequality to macroprudential tightening—but with wider confidence intervals and fewer statistically significant horizons. This lower precision is consistent with the shorter timeframe. At the same time, precision improves when the post-2008 estimations are conducted by country group, suggesting a trade-off between gauging the heterogeneity of effects across country types and the information loss from

reducing the sample size.

2.3.6 Discussion

Taken together, the results point to a robust negative relationship between macroprudential regulation and income inequality. The effect is present for both the market and disposable Gini measures, across the aggregate and disaggregated policy indices, and across different instrument classifications. Several patterns emerge consistently across the exercises.

First, the negative effect on inequality is primarily driven by tightening interventions. The tightening-only index yields effects that are quantitatively similar than those of the net aggregate index. The loosening-only index, while not statistically significant, produces point estimates that are broadly consistent with the overall pattern: loosenings tend to be associated with positive (inequality-increasing) responses, as one would expect if relaxing financial regulation allows the intermediation-driven inequality channel to operate more freely.

Second, the disposable Gini tends to respond more strongly and with greater statistical precision than the market Gini. This suggests that fiscal redistribution complements the inequality-reducing effects of prudential regulation, allowing for a clearer delimitation of the distributional consequences of financial oversight. In other words, the unintended inequality-reducing side effects of macroprudential tightenings appear to be reinforced—rather than offset—by the existing tax-and-transfer system.

Third, among the instrument classifications, the effects are strongest and most precisely estimated for lender-targeted, capital-based, resilience, and cycle-management tools. These are the instruments that most directly constrain the scale, leverage, and profitability of banking intermediation. Borrower-targeted instruments and tax-type tools, by contrast, produce insignificant or imprecisely estimated responses.

The estimated magnitudes should be interpreted as conservative. The impulse responses pool a large and heterogeneous set of economies—advanced, emerging, and developing—including countries with relatively low macroprudential interventionism, which attenuates the average effect. Moreover, the responses correspond to a single unit change in the policy index, that is, a one-instrument tightening in a given year. In practice, the macroprudential toolkit encompasses 17 instruments, and regulators frequently adjust several of them simultaneously, implying that the cumulative effect of a realistic policy package could be substantially larger.

These empirical findings suggest a plausible economic mechanism that we examine in later

sections. The intuition is that during periods of elevated financial intermediation, the benefits of banking activities—including lending spreads, capital gains, and profit distributions—accrue disproportionately to wealthier agents, such as bank shareholders and high-income entrepreneurs with greater access to financial markets. Macroprudential tightening curtails the scale and profitability of intermediation, thereby slowing this process. Motivated by this evidence, the model developed in the following sections incorporates macroprudential policy as a tax on banking intermediation returns, and features a production structure with capital-skill complementarity that generates heterogeneous income sources across agent types.

3 Model

To understand how macroprudential policy impacts income inequality, we build a closed-economy model with two types of infinitely lived agents: entrepreneurs and households. Entrepreneurs, who make up a $1-\mu$ share of the population, supply skilled labor to competitive firms and own the technology that transforms final goods into physical capital, which is financed through the banking sector. Households, comprising the μ share of the population, supply unskilled labor but lack a direct investment technology. Both agents can save in bank deposits and bonds.

To simplify the notation, we will consider the individual decisions for households and entrepreneurs in lowercase letters, and the rest of the variables for representative agents in uppercase letters.[9] We also indicate the relation between individual decisions and aggregate economic variables in the following subsections.

3.1 Households

The representative household solves,

$$\max_{\{c_{h,t},d_t,b_{h,t}\}} E_t \sum_{s=t}^{\infty} \beta^{s-t} \left[\frac{(c_{h,s})^{1-\gamma}}{1-\gamma} \right].$$

The budget constraint for the household is:

$$c_{h,t} + d_t + b_{h,t} + \frac{\eta_h}{2}(b_{h,t} - \bar{b}_h)^2 \leq w_{h,t}h_{h,t} + R_{b,t-1}b_{h,t-1} + R_{d,t-1}d_{t-1} + \pi_t, \tag{2}$$

[9]With the exception of gross interest rates that are also denoted with capital letters.

where $\beta \in (0,1)$ is the subjective discount factor, $c_{h,t}$ is household consumption, $\gamma > 0$ is the inverse of the intertemporal elasticity of substitution, and d_t is the amount of deposits held in the domestic bank with a gross rate of return $R_{d,t}$. An endowment of low-skilled labor is supplied inelastically at a wage $w_{h,t}$. The term π_t denotes the per-capita distribution of banking profits and the lump-sum fiscal component received by the household.

The household's optimizing decisions imply the Euler equations:

$$1 = \beta \mathbb{E}_t \left[R_{d,t} \left(\frac{c_{h,t+1}}{c_{h,t}} \right)^{-\gamma} \right], \tag{3}$$

$$1 + \eta_h (b_{h,t} - \bar{b}_h) = \beta \mathbb{E}_t \left[R_{b,t} \left(\frac{c_{h,t+1}}{c_{h,t}} \right)^{-\gamma} \right]. \tag{4}$$

3.2 Entrepreneurs

Entrepreneurs supply high-skilled labor $h_{e,t}$ inelastically to firms at wage rate $w_{e,t}$. They own a technology that transforms final consumption goods into physical capital k_t, which they supply to final good firms. Thus, they determine capital dynamics and investment flows. Moreover, the labor they provide is skilled in the sense that it is complementary to the capital stock. These features are reflected in the production function of the firms and in the sources of income of the entrepreneurs.

The representative entrepreneur's utility maximization problem is:

$$\max_{\{c_{e,t}, d_{e,t}, b_{e,t}\}} E_t \sum_{s=t}^{\infty} \beta^{s-t} \left[\frac{(c_{e,s})^{1-\gamma}}{1-\gamma} \right],$$

The entrepreneur funds consumption using skilled labor income, deposit and bond returns, and the profits from capital goods production and banking intermediation. The budget constraint is:

$$c_{e,t} + d_{e,t} + b_{e,t} + \frac{\eta_e}{2}(b_{e,t} - \bar{b}_e)^2 \leq w_{e,t} h_{e,t} + R_{b,t-1} b_{e,t-1} + R_{d,t-1} d_{e,t-1} + \pi_{e,t}.$$

The Euler equations with respect to the entrepreneur's choice variables, $c_{e,t}, d_{e,t}, b_{e,t}$, are:

$$1 = \beta \mathbb{E}_t \left[R_{d,t} \left(\frac{c_{e,t+1}}{c_{e,t}} \right)^{-\gamma} \right], \tag{5}$$

$$1 + \eta_e(b_{e,t} - \bar{b}_e) = \beta \mathbb{E}_t \left[R_{b,t} \left(\frac{c_{e,t+1}}{c_{e,t}} \right)^{-\gamma} \right]. \tag{6}$$

For convenience, we categorize the investment decision problem for this agent as that of a capital goods producer.[10]

3.2.1 Capital producers

Entrepreneurs also provide physical capital to the final goods firms. This implies that they determine the investment flows which are governed by the following dynamics:

$$k_t = i_t + (1 - \delta)\xi_t k_{t-1}, \tag{7}$$

where k_t is the physical capital that is determined in t and used for production in $t+1$, i_t is the investment flow in t, ξ_t is a capital quality factor, and δ is the depreciation rate parameter.[11]

The investment, at the same time, is decided in a competitive fashion by optimizing the profits of its provision, subject to adjustment costs with a size governed by the parameter ζ,

$$\max_{\{i_t\}} \mathbb{E}_t \sum_{s=0}^{\infty} \Lambda_{e,t|t+s} \left\{ Q_{t+s} i_{t+s} - i_{t+s} \left(1 + \frac{\zeta}{2} \left(\frac{i_{t+s}}{i_{t+s-1}} - 1 \right)^2 \right) \right\},$$

where $\Lambda_{e,t|t+1} = \beta \left(\frac{c_{e,t+1}}{c_{e,t}} \right)^{-\gamma}$ is the stochastic discount factor of the entrepreneur.

The FONC from this problem allows us to pin down the relative price of physical capital,

$$Q_t = 1 + \frac{\zeta}{2} \left(\frac{i_t}{i_{t-1}} - 1 \right)^2 + \zeta \left(\frac{i_t}{i_{t-1}} - 1 \right) \frac{i_t}{i_{t-1}} - \mathbb{E}_t \Lambda_{e,t|t+1} \zeta \left(\frac{i_{t+1}}{i_t} \right)^2 \left(\frac{i_{t+1}}{i_t} - 1 \right). \tag{8}$$

Aggregate variables dynamics. The relative population size of entrepreneurs implies that aggregate capital and investment are given by $K_t \equiv (1 - \mu)k_t$, $I_t \equiv (1 - \mu)i_t$.

[10] The results are analogous to those obtained when the investment decision is directly incorporated into the entrepreneur's utility maximization. However, the presentation is clearer due to the presence of adjustment costs in the investment goods units. For similar approaches involving this type of separation, see Gertler and Karadi (2011) and Banerjee et al. (2016), among others.

[11] The depreciation rate in this setup is not only δ but $1 - (1 - \delta)\xi_t$, which decreases with capital quality ξ_t.

3.3 Firms

The model features a continuum (measure one) of identical, competitive firms producing a homogeneous good (Y_t) with capital (K_{t-1}), unskilled labor ($H_{h,t}$), and skilled labor ($H_{e,t}$). Capital-skill complementarity is modeled using a nested CES production function, following Krusell, Ohanian, Ríos-Rull, and Violante (2000). Here, we assume that entrepreneurs provide both capital goods and the skilled labor needed to operate them.

$$Y_t = A_t \left[(\alpha_u)^{\frac{1}{\sigma}} (H_{h,t})^{\frac{\sigma-1}{\sigma}} + (1-\alpha_u)^{\frac{1}{\sigma}} [V_t]^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \tag{9}$$

where

$$V_t = \left[(\alpha_k)^{\frac{1}{\rho}} (\xi_t K_{t-1})^{\frac{\rho-1}{\rho}} + (1-\alpha_k)^{\frac{1}{\rho}} (H_{e,t})^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}. \tag{10}$$

In the above production structure, the parameters α_u and α_k determine the relative weight of low-skill labor and capital in the production process. The parameter $\rho > 0$ represents the elasticity of substitution between capital and high-skill labor, while the parameter $\sigma > 0$ indicates the elasticity of substitution between the capital-high-skill labor composite V_t and low-skill labor. Production exhibits capital-skill complementarity if $\sigma > \rho$. A_t is the aggregate productivity shock and ξ_t is the capital-quality term.

Firms maximize the following profit function given by the revenues from final goods production and undepreciated capital goods resales after production, minus the cost of labor inputs and the banking loans used for funding physical capital:

$$\max_{\{K_{t-1}, H_{h,t}, H_{e,t}\}} Y_t + (1-\delta)\xi_t Q_t K_{t-1} - w_{e,t} H_{e,t} - w_{h,t} H_{h,t} - R_{k,t} Q_{t-1} K_{t-1}.$$

Profit maximization implies the following factor demand functions for low-skill labor, high-skill labor, and capital:

$$w_{h,t} H_{h,t}^{\frac{1}{\sigma}} = A_t^{\frac{\sigma-1}{\sigma}} (\alpha_u Y_t)^{\frac{1}{\sigma}}, \tag{11}$$

$$w_{e,t} H_{e,t}^{\frac{1}{\rho}} = (1-\alpha_k)^{\frac{1}{\rho}} A_t^{\frac{\sigma-1}{\sigma}} ((1-\alpha_u) Y_t)^{\frac{1}{\sigma}} V_t^{\frac{\sigma-\rho}{\sigma\rho}}, \tag{12}$$

$$R_{k,t} = \frac{r_t + (1-\delta)\xi_t Q_t}{Q_{t-1}}, \tag{13}$$

where the marginal product of capital in the last expression is given by,

$$r_t = A_t^{\frac{\sigma-1}{\sigma}} \xi_t^{\frac{\rho-1}{\rho}} \left[(1-\alpha_u)Y_t\right]^{\frac{1}{\sigma}} \left(\frac{\alpha_k}{K_{t-1}}\right)^{\frac{1}{\rho}} V_t^{\frac{\sigma-\rho}{\sigma\rho}}. \tag{14}$$

Here, we assume that the capital is funded by selling company securities to banks in a one-to-one relationship. This implies that the marginal product of capital r_t can also be interpreted as the return from firms' securities.

The skill premium is calculated as:

$$\frac{w_{e,t}}{w_{h,t}} = \left(\frac{1-\alpha_u}{\alpha_u}\right)^{\frac{1}{\sigma}} \frac{H_{h,t}^{\frac{1}{\sigma}}}{H_{e,t}^{\frac{1}{\rho}}} (1-\alpha_k)^{\frac{1}{\rho}} \left[(\alpha_k)^{\frac{1}{\rho}} (\xi_t K_{t-1})^{\frac{\rho-1}{\rho}} + (1-\alpha_k)^{\frac{1}{\rho}} (H_{e,t})^{\frac{\rho-1}{\rho}}\right]^{\frac{\sigma-\rho}{\sigma\rho}}.$$

The skill premium expression makes clear that $w_{e,t}/w_{h,t}$ rises whenever capital deepens or capital quality improves ($\xi_t K_{t-1}$ increases), since the composite V_t enters with exponent $(\sigma-\rho)/(\sigma\rho) > 0$ under capital-skill complementarity ($\sigma > \rho$). This channel is central to the distributional consequences of macroprudential policy in the model. Since entrepreneurs supply high-skill labor and households supply low-skill labor, any policy that affects capital accumulation — by tightening or loosening credit intermediation — alters the skill premium and thereby shifts the relative labor incomes of the two groups. The Gini coefficient derived in Section 3.7 aggregates precisely this wage dispersion: a tighter macroprudential stance that curtails capital growth compresses the premium and reduces inequality, while a credit boom widens it. The quantitative strength of this link depends on the degree of capital-skill complementarity, which our calibration ($\sigma = 2.2 > \rho = 1.26$) places firmly in the range estimated for recent U.S. data.

3.4 Banks

The intermediation sector is modeled along the lines of Gertler and Karadi (2011), Agénor et al. (2021b), among others, with some modifications to account for the presence of entrepreneurs and the associated financial flows. We assume a continuous range of competitive banks, with a total measure of one. The typical bank accepts deposits from households (D_t) and entrepreneurs ($D_{e,t}$) at the deposit interest rate $R_{d,t}$. Here, $D_{e,t} \equiv (1-\mu)d_{e,t}$ is the total entrepreneurial deposits, and $D_t \equiv \mu d_t$ represents the total deposits held by households.

The representative bank maximizes its value function of financial intermediation:

$$J_{j,t}(N_{j,t}) = \max_{\{N_{j,t}, Z_{j,t}, D_t, D_{e,t}\}} \mathbb{E}_t \tilde{\Lambda}_{t|t+1} \left[(1-\theta) N_{j,t+1} + \theta J_{j,t+1} (N_{j,t+1}) \right],$$

where the stochastic discount factor is defined as a population-weighted average of the corresponding stochastic discount factors of households and entrepreneurs: $\tilde{\Lambda}_{t|t+1} = \mu \Lambda_{t|t+1} + (1-\mu)\Lambda_{e,t|t+1}$ with $\Lambda_{t|t+1} = \beta \left(\frac{c_{h,t+1}}{c_{h,t}} \right)^{-\gamma}$ and an analogous quantity for the entrepreneurs. $N_{j,t}$ is the net-worth of surviving banks. With probability $1-\theta$, the bank will fail and pay back its profits to its owners, and with probability θ the bank will be able to continue its business and pursue future profits.

The bank maximizes its value subject to the net worth dynamics, the incentive compatibility constraint, and the balance sheet constraint are satisfied:

$$\begin{aligned} N_{j,t+1} &= (1-\tau_{k,t+1}) R_{k,t+1} Q_t Z_{j,t} - R_{d,t}(D_t + D_{e,t}), \\ J_{j,t} &\geq \kappa Q_t Z_{j,t}, \\ Q_t Z_{j,t} &= N_{j,t} + D_t + D_{e,t}. \end{aligned}$$

The net worth equation incorporates the presence of potential macroprudential instruments charging the revenue of the bank. The ICC states that the bank's continuation value has to be greater than or equal to the potential value of defaulting, as the bank can run away with a portion κ of the claims on company securities.

In the Balance sheet, D_t and $D_{e,t}$ are the total deposits, $Q_t Z_{j,t}$ are the claims on capital stock sold by firms to banks in a one-to-one relationship, with $Z_{j,t} = K_{j,t}$. The left-hand side constitutes banks' assets, whereas the right-hand side comprises the banks' liabilities.

The optimality conditions are:

$$[Z_{j,t}] : \mathbb{E}_t \Omega_{t+1|t} \left((1-\tau_{k,t+1}) R_{k,t+1} - R_{d,t} \right) = \gamma_t \kappa, \tag{15}$$

$$[N_{j,t}] : j_t \left(1-\gamma_t\right) = \mathbb{E}_t \Omega_{t+1|t} R_{d,t}, \tag{16}$$

where γ_t is the Lagrange multiplier associated with the ICC and $\Omega_{t+1|t} = \Lambda_{t|t+1} \left(1 - \theta + \theta J_{t+1}\right)$ is the effective stochastic discount factor of the bank. In this problem, we should also note that the bank is only concerned with the aggregate deposits rather than with the distribution

of these between types (households versus entrepreneurs). The remaining post-equilibrium conditions below will account for this as well as for the symmetry across banks, and thus yield a remaining number of sector-level conditions:

$$j_t N_t = \kappa Q_t K_t, \tag{17}$$

$$N_t = \theta \left[(1 - \tau_{k,t}) R_{k,t} Q_{t-1} K_{t-1} - R_{d,t-1}(D_{t-1} + D_{e,t-1}) \right] + \delta_T Q_t K_{t-1}, \tag{18}$$

$$Q_t K_t = N_t + D_t + D_{e,t}. \tag{19}$$

We assume that the ICC binds henceforth and that the value of the bank can be expressed as the product of the marginal value of the bank j times the net worth: $J_{j,t} = j_t N_t$. Additionally, the second expression corresponds to the net worth of the entire financial system which is defined as θ times the net worth of individual surviving banks plus the start-up capital of new banks which is given by a small fraction δ_T times the value of the capital stock.[12]

3.5 Macroprudential policy

The macroprudential policy we consider in the model is a tax on the returns to banking intermediation, which is proportional to the gross returns times the value of intermediation in each period.[13]

The macroprudential regulator will run a balanced budget and rebate her revenue to households and entrepreneurs (in proportion to their population weights),

$$\tau_{k,t} Q_{t-1} R_{k,t} K_{t-1} + T_t = 0. \tag{20}$$

Policy setting. We consider a baseline policy rule for the macroprudential tax rate that sets the prudential instrument as a function of the deviation of the capital stock from its steady state. In later sections, we also compare other rules, including a case with an explicit inequality

[12] Similarly, these expressions make explicit the one-to-one relationship between the stock of assets intermediated and the stock of capital at the aggregate level that was alluded to when outlining the firm's problem, i.e., $Z_t = K_t$, where Z_t is the stock of securities from the firms.

[13] This policy instrument is set as a tax for simplicity, but it has been shown to be general enough to encompass the structure of a varied number of macroprudential policies (see for example Agénor et al., 2021a; Liu et al., 2023b; Aoki et al., 2016; Dennis and Ilbas, 2023). For a recent list of the types of instruments in the macroprudential toolkit, see Alam, Alter, Eiseman, Gelos, Kang, Narita, Nier, and Wang (2025)

target. Our baseline rule will also consider an inertia term:

$$\tau_{k,t} = \rho_\tau \tau_{k,t-1} + (1-\rho_\tau)\psi_\tau \left(\frac{K_t}{K^*} - 1\right) + \sigma_\tau \epsilon_{\tau,t}, \tag{21}$$

The parameter ρ_τ captures the inertia in the policy rule, while ψ_τ governs the response to deviations of the capital stock from its steady state level. Both parameters are calibrated in line with studies on macroprudential policy rules (e.g., Angelini, Neri, and Panetta, 2014; Hecker, Jang, Rubio, and Verona, 2024; Rubio and Carrasco-Gallego, 2015). Finally, we incorporate an i.i.d. policy shock $\epsilon_{\tau,t}$ with a standard deviation σ_τ.

3.6 Equilibrium

3.6.1 Exogenous processes

We allow for exogenous variation stemming from shocks to aggregate productivity and capital quality that follow autoregressive processes given by:

$$\ln A_t = \rho_A \ln A_{t-1} + \sigma_A \epsilon_{A,t}, \tag{22}$$

$$\ln \xi_t = \rho_\xi \ln \xi_{t-1} + \sigma_\xi \epsilon_{\xi,t}, \tag{23}$$

where a higher value of ξ implies a greater marginal return to physical capital and a lower depreciation rate. As in the literature, we interpret the latter shock as having a financial nature, given the mapping between the stock of assets and physical capital and its close link with banking intermediation profits.

3.6.2 Market clearing conditions and aggregation

The goods market clearing condition takes into account the relative population sizes of the agents, the bond adjustment costs, and the investment total cost (principal plus adjustment).

$$Y_t = \mu\left[c_{h,t} + \frac{\eta_h}{2}(b_{h,t} - \bar{b}_h)^2\right] + (1-\mu)\left[c_{e,t} + \frac{\eta_e}{2}(b_{e,t} - \bar{b}_e)^2\right] + (1-\mu) i_t \left(1 + \frac{\zeta}{2}\left(\frac{i_t}{i_{t-1}} - 1\right)^2\right). \tag{24}$$

Bonds are given in zero net supply, so that

$$\mu b_{h,t} + (1-\mu) b_{e,t} = 0. \tag{25}$$

Aggregation. As we have two types of agents (households with relative population size μ and entrepreneurs representing the rest), we must consider the aggregation relationships, which map their individual decisions into economy-wide variables.[14]

$$K_t = (1-\mu)k_t, \quad I_t = (1-\mu)i_t,$$

$$D_{e,t} = (1-\mu)d_{e,t}, \quad D_t = \mu d_t,$$

$$H_{h,t} = \mu h_{h,t}, \quad H_{e,t} = (1-\mu)h_{e,t}.$$

At the same time, we are considering a fixed labor supply, and thus it follows that $H_{h,t} = \mu$ and $H_{e,t} = (1-\mu)$.

3.6.3 Profit distribution

Banking profits and the lump-sum fiscal term are distributed across households and entrepreneurs in proportion to their population shares, while profits from capital-goods production accrue to entrepreneurs. Accordingly, aggregate profit distributions to households and entrepreneurs are given by

$$\Pi_t = \mu \left\{ (1-\theta) \left[(1-\tau_{k,t}) Q_{t-1} R_{k,t} K_{t-1} - R_{d,t-1} \widetilde{D}_{t-1} \right] - \delta_T Q_t K_{t-1} - T_t \right\}, \tag{26}$$

$$\begin{aligned} \Pi_{e,t} = (1-\mu) &\left\{ (1-\theta) \left[(1-\tau_{k,t}) Q_{t-1} R_{k,t} K_{t-1} - R_{d,t-1} \widetilde{D}_{t-1} \right] - \delta_T Q_t K_{t-1} - T_t \right\} \\ &+ \left\{ Q_t I_t - I_t \left[1 + \frac{\zeta}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \right] \right\}, \end{aligned} \tag{27}$$

where $\widetilde{D}_t \equiv D_t + D_{e,t}$ denotes aggregate deposits. The corresponding per-capita profit distributions entering the household and entrepreneur budget constraints are defined by

$$\Pi_t = \mu \pi_t, \qquad \Pi_{e,t} = (1-\mu)\pi_{e,t}. \tag{28}$$

3.6.4 Model equilibrium

Given a set of macroprudential policies $\{\tau_{k,t}\}$, the tax-distorted competitive equilibrium for this economy consists of allocations for households $\{c_{h,t}, d_t, b_{h,t}\}$, entrepreneurs $\{c_{e,t}, d_{e,t}, b_{e,t}, i_t, k_t\}$, firms $\{Y_t, K_t, H_{h,t}, H_{e,t}, V_t\}$, and banks $\{N_t, D_t, D_{e,t}\}$, along with prices $\{w_{h,t}, w_{e,t}, r_t, R_{d,t}, R_{b,t},$

[14] Some of these relationships are mentioned in the above agents' subsections for clarity.

$R_{k,t}, Q_t\}$, aggregate variables $\{I_t, T_t\}$, auxiliary variables $\{j_t, \gamma_t\}$, and exogenous state variables $\{A_t, \xi_t\}$, such that:

- Households maximize utility subject to their budget constraint.
- Entrepreneurs maximize utility (with budget constraint implicitly satisfied by Walras' Law).
- Capital producers maximize profits subject to adjustment costs.
- Firms maximize profits.
- Banks maximize franchise value subject to ICC, balance sheet, and net worth dynamics.
- Government budget balances.
- All markets clear: goods, labor, bonds, and deposits.
- Aggregation relationships hold.
- Exogenous state variables evolve according to their autoregressive processes.

The system consists of 29 equations determining 29 endogenous variables: equations (2)–(20) and (22)–(25), together with the six aggregation relationships in Section 3.6.2 and the profit terms in (26)–(28), which enter the budget constraints. Under a policy rule such as (21), $\tau_{k,t}$ becomes endogenous, adding one equation and one variable. When we introduce inequality to the model, the Gini coefficient enters the system as explained in Section 3.7.

3.7 Gini coefficient

To evaluate the distributional consequences of macroprudential policy, the model requires a measure of inequality that links the underlying income components to an aggregate statistic. This subsection derives the model-implied Gini coefficient and expresses it in terms of entrepreneurial and household incomes. The resulting expression provides the key object used in the quantitative policy analysis of Section 5.

We measure inequality with the Gini coefficient on total per-capita income of the two agent types. Entrepreneur and household incomes combine capital and labor earnings:

$$y_{e,t} = w^k_{e,t} + w^l_{e,t}, \qquad y_{h,t} = w^k_{h,t} + w^l_{h,t}.$$

Consistent with the budget constraints mentioned before, we define capital income as the net return on financial assets together with the per-capita profit and transfer term received by each type:

$$w^k_{e,t} = (R_{d,t-1} - 1)d_{e,t-1} + (R_{b,t-1} - 1)b_{e,t-1} + \pi_{e,t},$$
$$w^k_{h,t} = (R_{d,t-1} - 1)d_{t-1} + (R_{b,t-1} - 1)b_{h,t-1} + \pi_t,$$

while labor income is

$$w^l_{e,t} = w_{e,t}h_{e,t}, \qquad w^l_{h,t} = w_{h,t}h_{h,t}.$$

Here, π_t and $\pi_{e,t}$ are the per-capita profit and transfer terms that already incorporate each group's share of banking profits and the fiscal component of the macroprudential tax. The entrepreneur term $\pi_{e,t}$ additionally includes profits from capital-goods production. Hence no separate banking-profit term is required in the income definitions.

With population shares μ for households and $1 - \mu$ for entrepreneurs, the two-group Gini coefficient is

$$\text{Gini}_t = \frac{\mu(1-\mu)\,|y_{e,t} - y_{h,t}|}{\mu y_{h,t} + (1-\mu)y_{e,t}}. \tag{29}$$

This measure captures both sources of income dispersion in the model. The first is the skilled–unskilled wage differential generated by capital–skill complementarity. The second is the difference in financial income arising from the two groups' deposit and bond positions and from their respective profit and transfer terms. Macroprudential policy can therefore affect inequality through both labor-income and financial-income channels.

Appendix B.1 shows that, at the deterministic steady state, this expression reduces to a particularly transparent form. The budget constraints imply $y_e = c_e$ and $y_h = c_h$. Defining $\omega \equiv c_e/c_h > 1$, the Gini can therefore be written as

$$G = \frac{\mu(1-\mu)(\omega - 1)}{\mu + (1-\mu)\omega}.$$

Equivalently, defining the entrepreneur income share

$$\Lambda_e \equiv \frac{(1-\mu)y_e}{\mu y_h + (1-\mu)y_e} = \frac{(1-\mu)c_e}{\mu c_h + (1-\mu)c_e},$$

the Gini satisfies the income-share identity (B.1):

$$G = \Lambda_e - (1 - \mu).$$

The Gini therefore equals the gap between the entrepreneurs' income share and their population share. This representation provides the distributional object used in the policy analysis below, while Appendix B.1 gives the full steady-state derivation.

This expression also clarifies how macroprudential policy affects inequality in steady state. Under the steady-state conditions considered in Appendix B, common nonlabor income components cancel from the income difference, so that inequality can be expressed in terms of the skilled–unskilled wage gap relative to aggregate disposable income. A macroprudential tightening reduces the steady-state capital stock and, through capital–skill complementarity, compresses this wage gap. Proposition B.1 shows that the Gini declines locally when the wage-gap compression is sufficiently strong relative to the change in aggregate disposable income, a condition that is satisfied at the baseline calibration.

4 Calibration

The model is calibrated at a quarterly frequency to match key moments of the U.S. economy. Table 3 in Appendix C summarizes all parameter values; here we describe the main choices.

Standard parameters. We set $\beta = 0.99$, $\gamma = 2$, and $\delta = 0.025$ (10% annual depreciation), following standard practice in the quarterly DSGE literature. The population share of households $\mu = 0.55$ matches Li et al. (2023), who distinguish between skilled (entrepreneur) and unskilled (household) worker shares based on U.S. occupation data.

Production and capital-skill complementarity. The nested CES production structure follows Krusell et al. (2000). We set the outer substitution elasticity $\sigma = 2.2$ and the inner elasticity $\rho = 1.26$ based on the estimates of Duffy et al. (2004), which implies capital-skill complementarity since $\sigma > \rho$. Given these elasticities, the share parameters (α_u, α_k) are calibrated jointly to match two empirical targets: (i) a skill premium $w_e/w_h = 1.5$, consistent with Papageorgiou and Chmelarova (2005) and Bureau of Labor Statistics CPS data, and (ii) a capital income share $rK/Y = 0.33$, from Penn World Tables and Karabarbounis and Neiman (2014). The solution yields $\alpha_u = 0.6048$ and $\alpha_k = 0.2276$.

Banking sector. The banking block is calibrated following Gertler and Karadi (2011). The banker survival probability $\theta = 0.972$ corresponds to an average banker horizon of approximately 36 quarters (9 years), consistent with their reported bank holding-company data. Given θ, we solve analytically for (κ, δ_T) to jointly target (i) a bank leverage ratio $K/N = 4$, taken from the Federal Reserve Financial Accounts (Flow of Funds Z.1 release), and (ii) an annual credit spread $(R_k - R_d) \times 400 = 1$ (i.e., 100 basis points), consistent with the Moody's Baa–10yr Treasury spread (FRED series BAA10Y) and the Gilchrist and Zakrajšek (2012) (GZ) spread. This gives $\kappa = 0.3847$ and $\delta_T = 0.00212$.

Other parameters. The TFP shock process parameters ($\rho_A = 0.85$, $\sigma_A = 0.007$) are set as in Smets and Wouters (2007), and the capital quality ones ($\rho_\xi = 0.85$, $\sigma_\xi = 0.005$) follow Gertler and Karadi (2011). The policy-rule parameters (ρ_τ, ψ_τ, ψ_K, ψ_G) are design choices rather than structural targets; we set $\rho_\tau = 0.65$, $\psi_\tau = \psi_K = 0.75$, and $\psi_G = 0.25$, consistent with the moderate inertia used in studies of macroprudential rules with an active fundamental component (Angelini et al., 2014; Rubio and Carrasco-Gallego, 2015).

5 Simulation results under alternative policy frameworks

This section evaluates the dynamic implications of alternative macroprudential policy rules under three structural disturbances: a total factor productivity (TFP) shock, a capital quality shock, and an exogenous macroprudential policy shock. The objective is not only to compare the stabilization properties of alternative rules, but also to identify the transmission channels through which macroprudential regulation affects both aggregate activity and the distribution of income.

The five policy specifications are summarized in Table 1. In Specification 1 the macroprudential tax consists of a one-shot innovation without rule-induced endogenous feedback. Specifications 2–4 are instrument-based rules that adjust the macroprudential tax in response to lending, output, or a combination of lending and inequality. Specification 5 differs conceptually by targeting an endogenous financial quantity—bank leverage—rather than macroeconomic outcomes directly. This distinction proves important for the propagation of shocks.

Table 1: Macroprudential Policy Specifications

Specification	Description	Policy Rule
1	One-shot (no rule)	$\tau_{k,t} = \sigma_\tau \epsilon_{\tau,t}$
2	Lending rule	$\tau_{k,t} = \rho_\tau \tau_{k,t-1} + (1-\rho_\tau)\psi_\tau(\frac{K_t}{K^*} - 1) + \sigma_\tau \epsilon_{\tau,t}$
3	GDP rule	$\tau_{k,t} = \rho_\tau \tau_{k,t-1} + (1-\rho_\tau)\psi_\tau(\frac{Y_t}{Y^*} - 1) + \sigma_\tau \epsilon_{\tau,t}$
4	Gini-augmented: Lending and Gini	$\tau_{k,t} = \rho_\tau \tau_{k,t-1} + (1-\rho_\tau)\left[\psi_K(\frac{K_t}{K^*} - 1) + \psi_G(\frac{G_t}{G^*} - 1)\right] + \sigma_\tau \epsilon_{\tau,t}$
5	Leverage rule	$\tau_{k,t} = \rho_\tau \tau_{k,t-1} + (1-\rho_\tau)\psi_\tau(j_t/j^* - 1) + \sigma_\tau \epsilon_{\tau,t}$

Notes: $\rho_\tau = 0.65$ for all active rules. Starred variables denote steady-state values. G denotes the Gini coefficient.

5.1 Responses to a TFP shock

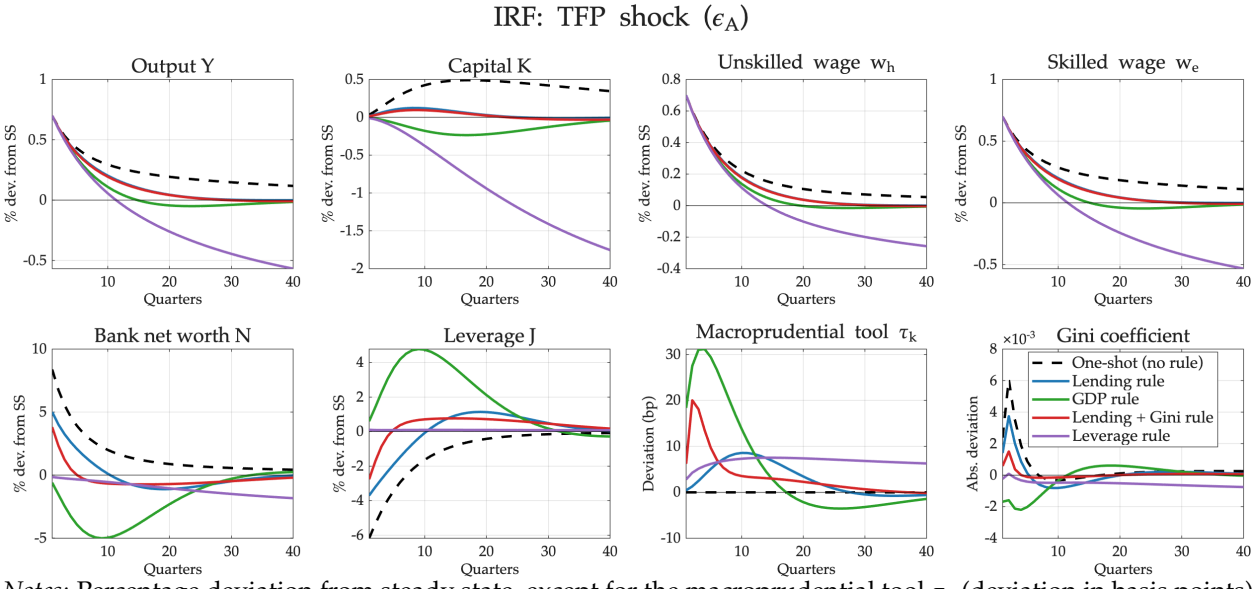
Figure 5 reports impulse responses to a one-standard-deviation positive TFP shock ($\sigma_A = 0.007$). The productivity improvement raises output and wages on impact across specifications. Because the disturbance originates on the production side of the economy rather than in the financial sector, the Gertler–Karadi amplification channel plays only a limited role in its propagation, so the aggregate responses are smaller and less persistent than those to the capital quality shock examined below. The endogenous feedback embodied in the active rules therefore has limited scope to reshape the transmission of the shock to output and wages. Its imprint is nonetheless visible in capital accumulation, where the specifications differ notably.

Some of the instrument-based rules (Specifications 2–3) dampen the investment and capital responses relative to the no-rule setup by tightening the macroprudential stance as activity expands: the lending rule preserves a smaller positive capital hump, while the GDP rule undershoots the capital steady state. A salient difference between these rules is that the lending rule can only respond with a delay, since the prudential tool is set as a function of the predetermined capital stock. By contrast, the GDP rule responds immediately. As a consequence, the GDP-based rule displays a markedly stronger countercyclicality while the lending-based rule tightening is delayed and more gradual. The other rules also differ in their implied capital accumulation dynamics. The Gini-augmented rule (Specification 4) tightens sharply on impact, responding to the initial rise in inequality, but relaxes the stance quickly as the Gini recedes, yielding a capital path that resembles that of the lending rule. The leverage rule (Specification 5), by contrast, shows a persistent tightening pattern, leading to

the strongest contraction of capital across all regime setups.

Higher productivity also affects the income distribution, raising entrepreneurs' capital-goods profits on impact and, as capital accumulates, widening the skilled–unskilled wage differential through capital–skill complementarity. As a result, the Gini coefficient rises on impact in the absence of a systematic rule. The increase is largest under the one-shot policy, where no countercyclical feedback offsets it, and is contained by the rules leaning against the expansion. The GDP rule even reverses it during the first years after the shock. Because output jumps on impact, this rule tightens immediately and forcefully, whereas the lending rule reacts only as capital gradually accumulates. The early tightening raises the tax revenue rebated to both groups, which lifts aggregate income relative to the income gap, and also moderates capital accumulation, thereby dampening the wage gap expansion. Absent this redistribution of the tax revenue, the wage-gap effect would dominate, and the Gini would rise in the first quarters—a pattern that echoes the stronger inequality-reducing response of the disposable Gini, measured after taxes and transfers, in Section 2. The leverage rule also keeps the Gini near or below its steady state, but at the cost of a persistent contraction of capital and output. Notably, the rules without an explicit inequality target can still reduce the Gini increase non-trivially via their countercyclical effect on the business cycle, intermediation, and skill premium.

Figure 5: Impulse Responses to a TFP Shock (ε_A)



Notes: Percentage deviation from steady state, except for the macroprudential tool τ_k (deviation in basis points) and the Gini coefficient (absolute deviation). Black dashed: One-shot (no rule). Blue: Lending rule. Green: GDP rule. Red: Lending + Gini rule. Purple: Leverage rule. One-standard-deviation shock ($\sigma_A = 0.007$).

5.2 Responses to a capital quality shock

Figure 6 reports impulse responses to a positive capital quality shock ($\sigma_\xi = 0.005$). Unlike the TFP disturbance, this shock operates directly through banks' balance sheets, and therefore it activates the financial amplification mechanism more strongly. Improvements in capital quality increase asset values, strengthen bank net worth, expand intermediation, and support a persistent increase in lending, investment, and output. Consequently, the differences across policy rules become considerably more pronounced.

The one-shot (no systematic rule) specification exhibits the strongest expansion in credit and capital accumulation. The lending and GDP rules lean against the financial expansion by tightening the macroprudential tax as lending and output rise, thereby moderating the associated investment increase and reducing the persistence of the boom.

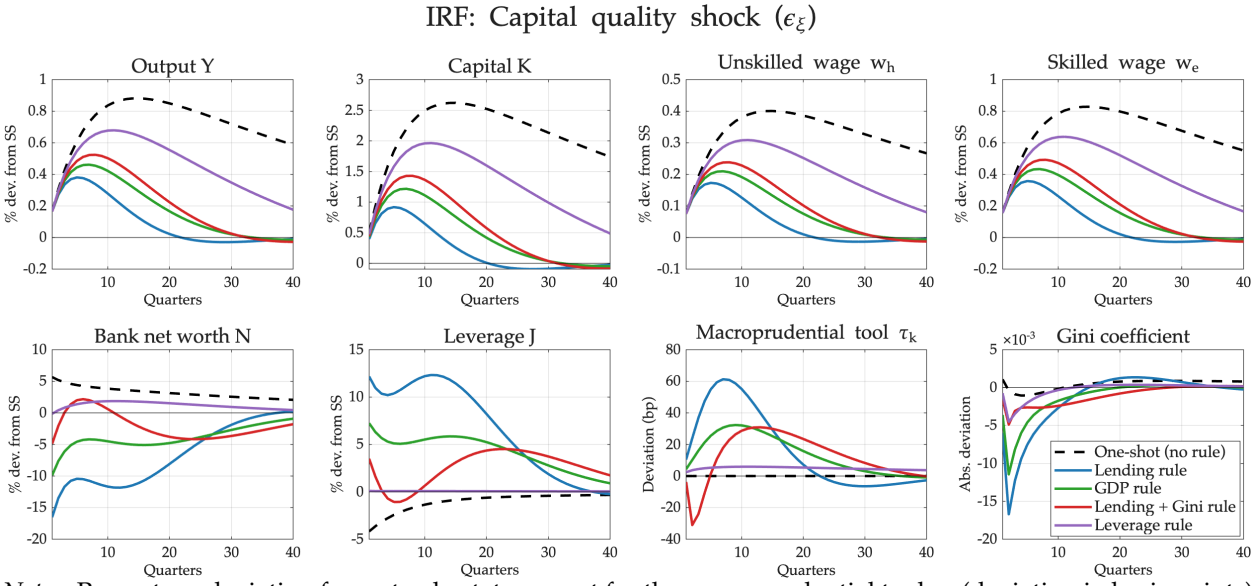
The greatest heterogeneity across specifications emerges in the distributional responses, and the pattern is more subtle than a usual ranking of the rules would suggest. Absent any rule, the Gini barely moves or slightly lowers. The lending and GDP rules, by contrast, deepen the decline with further tightening interventions. The Gini-augmented rule, in turn, penalizes the decrease in the Gini that the expansion and its lending component would otherwise induce. It initially loosens the instrument, a de facto intermediation subsidy, and returns to taxing intermediation later, as lending expands, but by less than the lending rule because the Gini is still below its steady state, which exerts a partially countervailing subsidizing force. This rule then generates a more stable Gini path than most other systematic rules, which in the case of inequality itself is not particularly noticeable—the Gini coefficient is already a relatively stable variable, and its policy-related emphasis hinges on potential reductions in its level—and instead can be interpreted as a negation of the inequality-reducing effects from macroprudential oversight.

The leverage rule, by contrast, generates almost no policy response. This rule has limited countercyclical properties by construction, as it targets a ratio whose components move together: the capital quality shock raises the nominal value of assets and bank net worth simultaneously, thereby mitigating the implied extent of the intervention, keeping the economy closer to the no-rule regime, and leading to weaker equality gains than the lending and GDP rules. This illustrates how macroprudential policies do not align by default with the distributional objectives of an economy.

Finally, a further pattern emerges from a reverse reading of the productivity and financial

shocks—i.e., during downturns. In the (one-shot) no-rule setup, leverage rises with no correction mechanism, whereas the lending, GDP, and leverage rules mitigate it by containing the increase, inducing a deleveraging, or holding leverage at a target. The exception is the Gini-augmented rule, which penalizes any movement in inequality, reductions included, casting further doubts on the desirability of such a setup.

Figure 6: Impulse Responses to a Capital Quality Shock (ε_ξ)



Notes: Percentage deviation from steady state, except for the macroprudential tool τ_k (deviation in basis points) and the Gini coefficient (absolute deviation). Black dashed: One-shot (no rule). Blue: Lending rule. Green: GDP rule. Red: Lending + Gini rule. Purple: Leverage rule. One-standard-deviation shock ($\sigma_\xi = 0.005$).

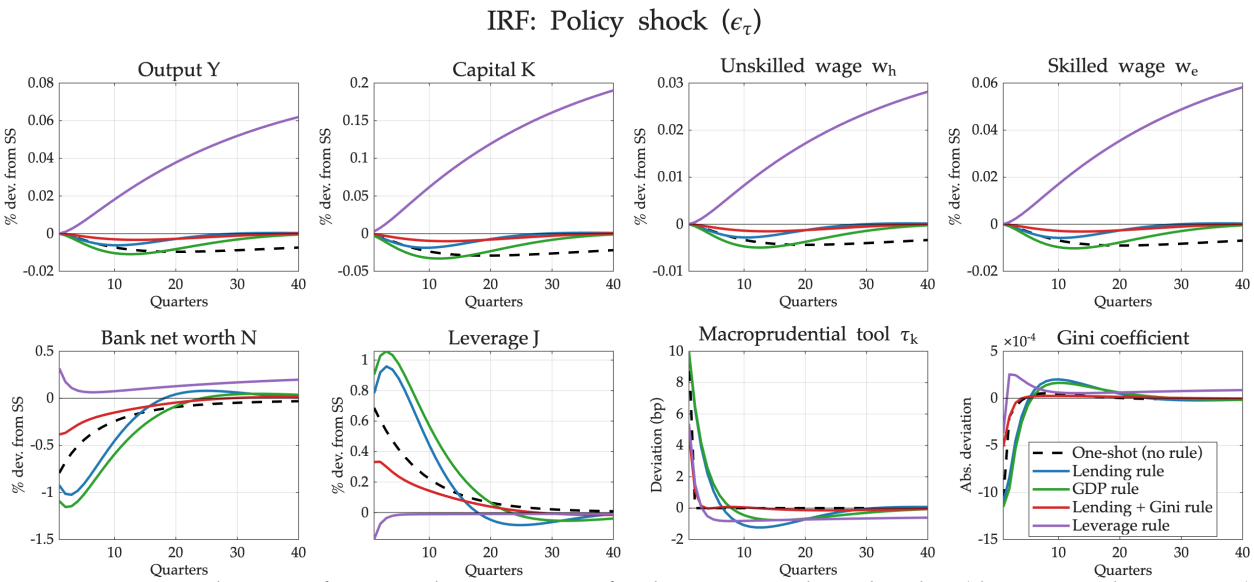
5.3 Responses to a policy shock

Figure 7 reports impulse responses to an exogenous macroprudential policy shock ($\sigma_\tau = 0.001$). Relative to the previous experiments, the quantitative effects are considerably smaller, reflecting that an unexpected policy tightening represents a weaker disturbance than either a productivity shock or a financial shock, which, by contrast, are expected to be fundamental drivers of capital accumulation and thus financial intermediation. Accordingly, this experiment is best interpreted as illustrating the propagation properties of alternative policy rules rather than their contribution to business-cycle stabilization.

Without a systematic rule, the higher macroprudential tax reduces lending, investment and output through tighter financial conditions. The instrument-based rules partially offset this contraction in subsequent periods because weaker lending and output automatically induce a relaxation of the policy instrument. Consequently, the active rules generate smoother

adjustment paths than the one-shot exogenous policy setup.

Figure 7: Impulse Responses to a Policy Shock (ε_τ)



Notes: Percentage deviation from steady state, except for the macroprudential tool τ_k (deviation in basis points) and the Gini coefficient (absolute deviation). Black dashed: One-shot (no rule). Blue: Lending rule. Green: GDP rule. Red: Lending + Gini rule. Purple: Leverage rule. One-standard-deviation shock ($\sigma_\tau = 0.001$).

The leverage-based rule is the exception to this pattern, and the contrast can be instructive. Under this specification the tightening shock becomes expansionary. Output, capital, and wages rise. The reason is that leverage falls, prompting the rule to relax the instrument and subsidize intermediation. The mechanism in this case works through the balance sheet of the banks: as the capital stock is predetermined, higher anticipated prices of capital lead to an increase in net worth, which in turn lowers the leverage ratio—the capital revaluation here raises net worth proportionally more than the value of the bank assets and therefore dominates the response of leverage, in accordance with the amplification mechanism mentioned in Gertler and Karadi (2011), whose framework we adapt. As a result, intermediation expands, and together with the economic expansion, the Gini coefficient reverses its initial decline and overshoots its steady state from the second quarter onward. Once again, the leverage rule generates a distributional behavior contrary to that of the other standard rules with countercyclical properties, and acts in opposition to the bulk of our empirical evidence on a negative correlation between macroprudential policy and inequality.[15]

[15]That said, this can also explain why some studies in the literature, and even some of our results (for borrowers), may point to a positive feedback between regulation and inequality for some specific instruments. In our estimates, the negative result is captured by the capital-based classification, which does include a leverage requirement. The majority of the instruments in that category, however, are capital-adequacy tools with countercyclical properties.

5.4 Income sources and the role of redistribution

We can exploit the additive nature of the subcomponents of the Gini coefficient to shed further light on the responses above by decomposing the Gini IRF into the individual contributions of the income sources. We do this exercise in Appendix D. Figures 17–19 report the results for each shock and rule. Even though this is an accounting exercise that only complements the IRF results, it reveals additional links between our model and the empirical evidence. In particular, it highlights the role of redistribution in the inequality response to policy.

In our model, the tax revenues are rebated to both types of agents in equal per-capita terms, and thus a tightening lifts aggregate income relative to the income gap and lowers the Gini. This channel accounts for a sizable fraction of the responses for some shocks and can deepen the inequality decline of the tightenings, especially under the countercyclical rules (lending and GDP), which tighten the most as lending expands. Conversely, when a rule relaxes the tool, the rebate turns into a lump-sum tax, contributing to the Gini overshoot under the leverage rule. These patterns connect with the stronger inequality-reducing effects on the disposable Gini shown in Section 2, suggesting that, as in the data, the equalizing role of regulation is not limited to its direct effect on intermediation but also operates through redistribution.

A further feature worth noting is that the wage-gap channel builds gradually as capital accumulates, in line with its central role in the analytical results (Appendix B), and the countercyclical rules dampen it precisely by moderating capital accumulation. Finally, a caveat is in order: the income contributions are not orthogonal. The same tightening that raises the rebate also curbs intermediation and bank payouts, pushing inequality in the opposite direction. The decomposition is therefore best read as indicative evidence that redistribution can matter, rather than as a quantitative assessment of each channel.

5.5 Implications for macroprudential policy design

Taking stock, several lessons emerge. The ranking of macroprudential rules depends on the transmission channel of the disturbance rather than on any intrinsic dominance of a particular alternative. First, following a TFP shock the differences across rules are limited for output and wages—because the expansion originates on the real side and financial amplification is secondary—though they remain visible in capital accumulation, where the lending rule, by virtue of its dependence on a predetermined variable, yields a delayed and more gradual coun-

In contrast, borrower-targeting instruments include only ratio-based requirements—closer in nature to the leverage ratio—and in some cases can even display a positive, albeit non-significant, correlation.

tercyclical impact than an output-based rule, albeit with longer-lasting tightenings. Second, under the capital quality shock, financial amplification becomes central, and the rules separate more notably: the lending and GDP rules lean most forcefully against the credit and capital boom, while the leverage rule keeps bank net worth closest to its steady state.

Turning to the distributional implications, the results caution against the conventional intuition that could motivate the addition of explicit inequality objectives. Rules leaning against the cycle already reduce inequality as a by-product to a substantial extent. The Gini-augmented rule, by contrast, does not enhance the distributional benefits, and instead forfeits most of these because it prioritizes a stabilization of the Gini coefficient over its reduction, showing that explicit distributional objectives (in a policy rule) might not be necessary, nor even desirable. The policy shock, though smaller in magnitude, is informative about propagation: it isolates a feedback specific to the leverage rule, whereby a fall in leverage induces an automatic loosening of the instrument that even reverses the intended tightening—and the sign of the distributional response. This is a property worth bearing in mind when targeting endogenous financial quantities. Overall, standard instrument-based rules—with no explicit equality targeting—deliver the most consistent stabilization profile across the three shocks and, as a by-product, can reduce inequality in the short run.

6 Conclusion

While the effectiveness and macroeconomic consequences of macroprudential regulation have been studied extensively, its distributional impact and the mechanisms involved remain a relatively unexplored area of research. We explore this dimension on two fronts. First, we estimate lag-augmented local projections on a panel of advanced, emerging, and developing economies over 1999–2024, and find that macroprudential tightenings reduce income inequality gradually rather than on impact, for both market and disposable Gini coefficients, with a stronger effect on the latter. Our estimates, based on a recent and rich dataset of prudential interventions, are informative about the direction of the effects and their location within the toolkit. The effect is concentrated in instruments targeting lenders—capital-based, resilience, and cycle-management tools—while borrower-targeting instruments do not display the same relationship.

To account for these patterns, we develop a model in which banks intermediate under agency frictions and production features capital–skill complementarity. In our framework, a macroprudential tightening restrains financial intermediation and capital accumulation

and, through capital–skill complementarity, compresses the wage gap between skilled and unskilled workers. We show analytically that inequality falls when this compression is strong relative to the response of aggregate disposable income, a condition that holds amply in our calibration. Because the mechanism operates through the lender's balance sheet, it is consistent with our empirical finding that the inequality effects of prudential interventions are concentrated among lender-targeted instruments.

We then evaluate a set of alternative policy rules and find that rules leaning against the financial cycle can reduce inequality as a by-product, whereas explicitly incorporating inequality as an additional policy target does not necessarily strengthen these distributional gains. A Gini-augmented rule can, in fact, forfeit most of them, as stabilizing inequality may call for instrument adjustments that run counter to those warranted by financial conditions. Rules targeting more complex financial quantities with weaker countercyclical properties can also generate unintended consequences. In particular, under a leverage-based rule, the policy instrument can loosen endogenously following an exogenous tightening, to the point of reversing the intended contraction and raising inequality, which may help explain why some studies find a positive relationship between prudential regulation and inequality. Overall, our results suggest that financial-stability and inequality objectives need not be in conflict, but that their alignment depends on the choice and design of the macroprudential policy rule.

Future research could explore the distributional effects of financial regulation in open-economy setups. Liu, Spiegel, and Zhang (2023a) already indicate that capital inflows can worsen inequality. A natural question then is whether prudential interventions remain effective in reducing inequality in an environment in which external financial conditions also shape the income distribution.

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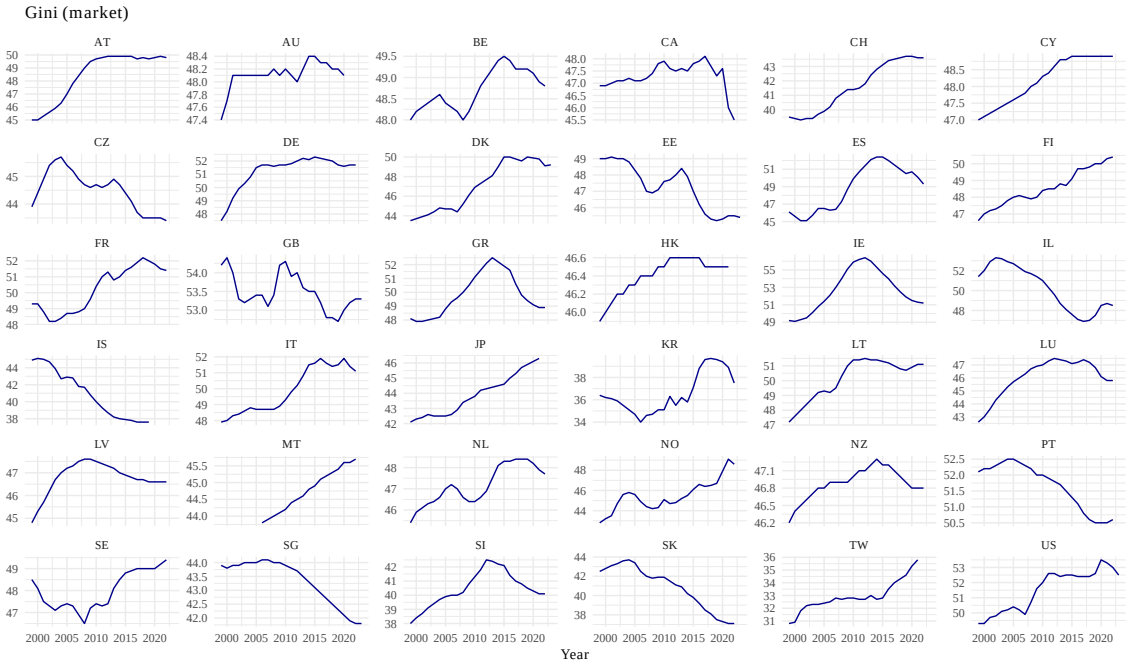
A Additional information on the data sample

This appendix presents additional descriptive information on the main variables used in the empirical analysis. In particular, we display the time-series dynamics of the Gini coefficients and macroprudential policy stances by country.

Table 2: Data Description and Sources

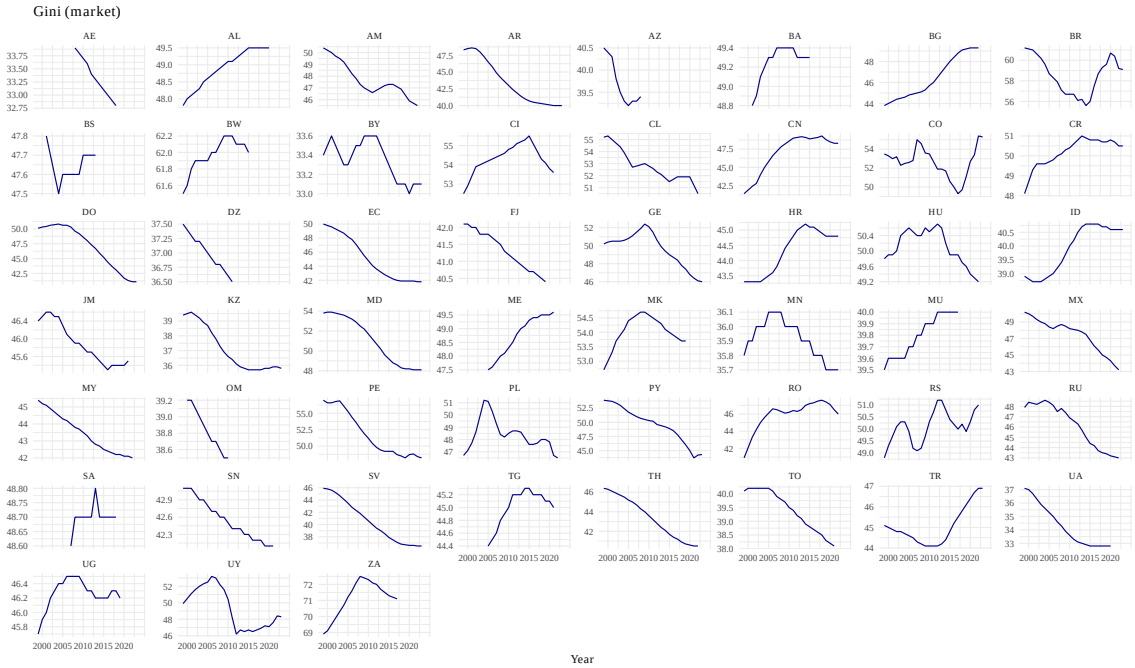
Variable	Description	Source
Dependent variables		
Gini (market)	Gini coefficient of market income	SWIID
Gini (disposable)	Gini coefficient of disposable income	SWIID
Macroprudential policy		
MaP (all)	Sum of tightenings and loosenings across 17 instruments	iMaPP–IMF
MaP (tightenings)	Sum of tightenings only	iMaPP–IMF
MaP (loosenings)	Sum of loosenings only	iMaPP–IMF
Control variables		
Banking crisis dummy	Indicator for banking crisis (5-year window)	Global Macro Database
Real GDP growth	Log-difference of real GDP	Global Macro Database
Inflation	CPI inflation rate	Global Macro Database
Central bank rate	Policy interest rate	Global Macro Database
Capital inflows/GDP	Gross capital inflows as share of GDP	IMF–IFS & GMD
Domestic credit/GDP	Private credit by deposit money banks (% GDP)	World Bank WDI
Financial development	IMF Financial Development Index (log-diff.)	IMF
Exchange rate depr.	Log-difference of nominal ER (vs. USD)	Global Macro Database

Figure 8: Market Gini Coefficient by Country - Advanced Economies



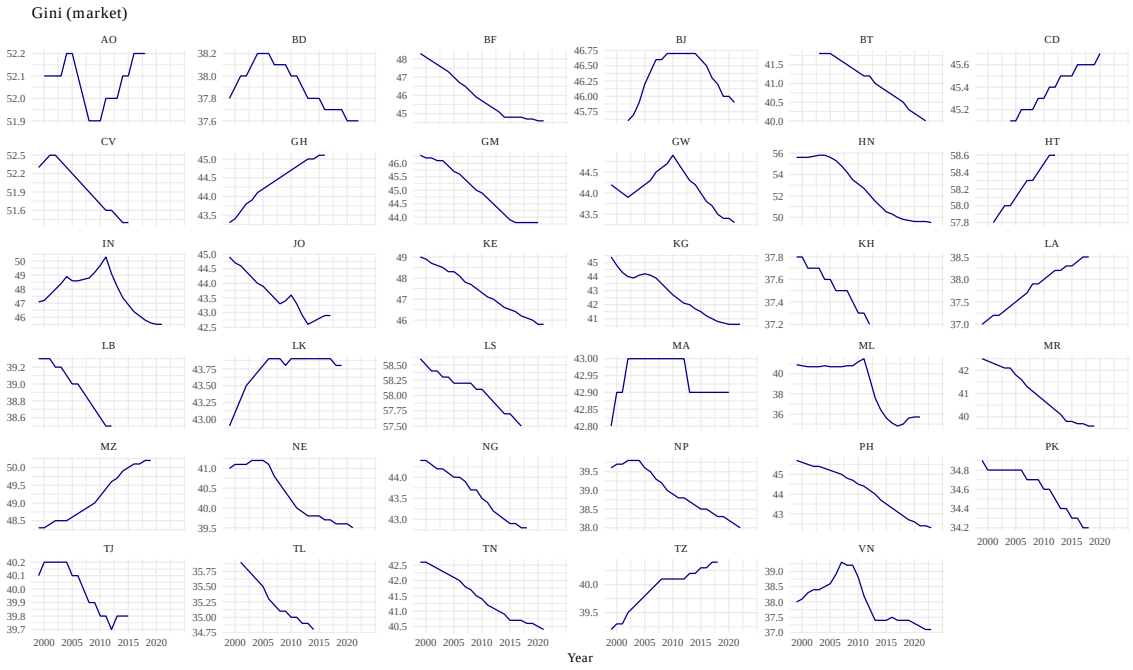
Notes: Data source: Standardized World Income Inequality Database (SWIID).

Figure 9: Market Gini Coefficient by Country - Emerging Markets



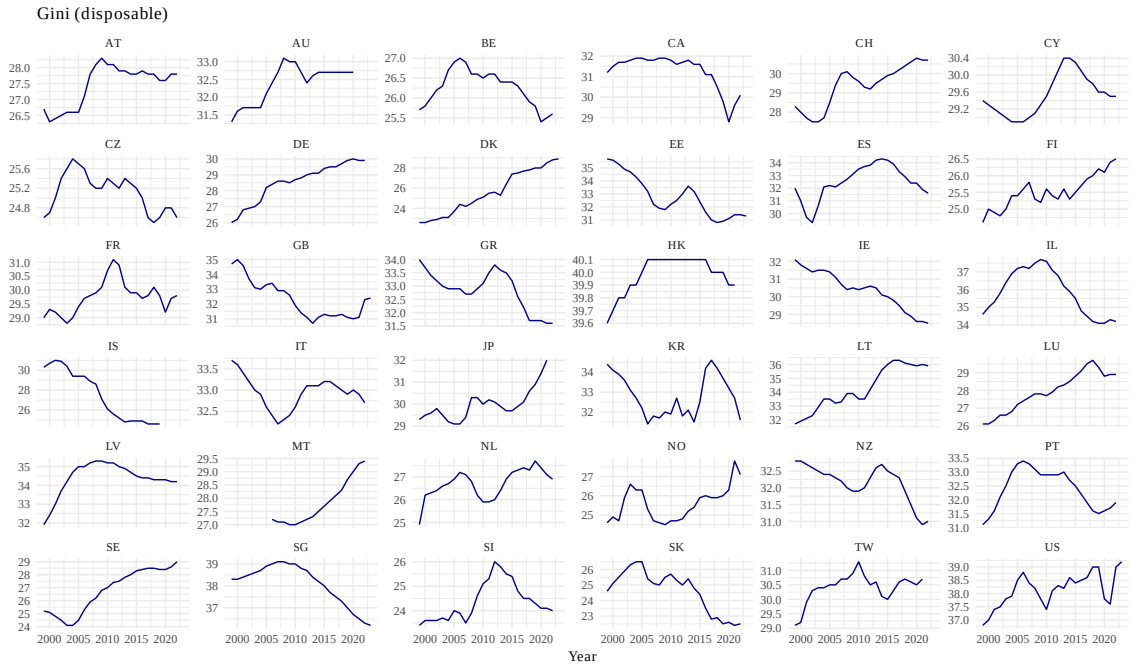
Notes: Data source: Standardized World Income Inequality Database (SWIID).

Figure 10: Market Gini Coefficient by Country - Developing Economies



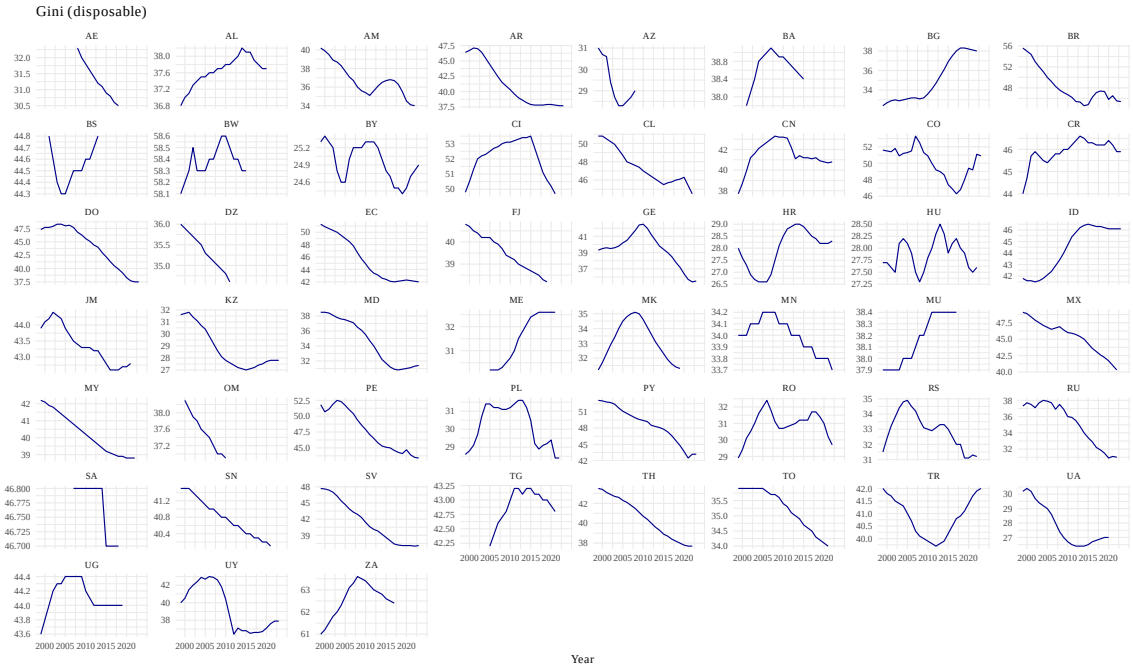
Notes: Data source: Standardized World Income Inequality Database (SWIID).

Figure 11: Disposable Gini Coefficient by Country - Advanced Economies



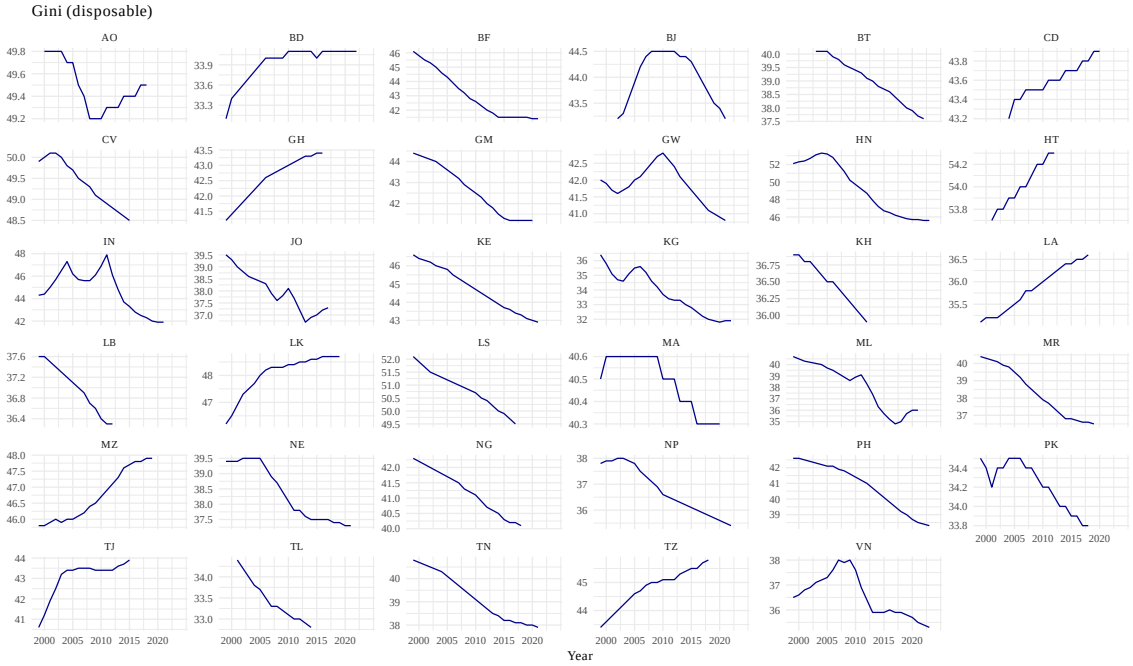
Notes: Data source: Standardized World Income Inequality Database (SWIID).

Figure 12: Disposable Gini Coefficient by Country - Emerging Markets



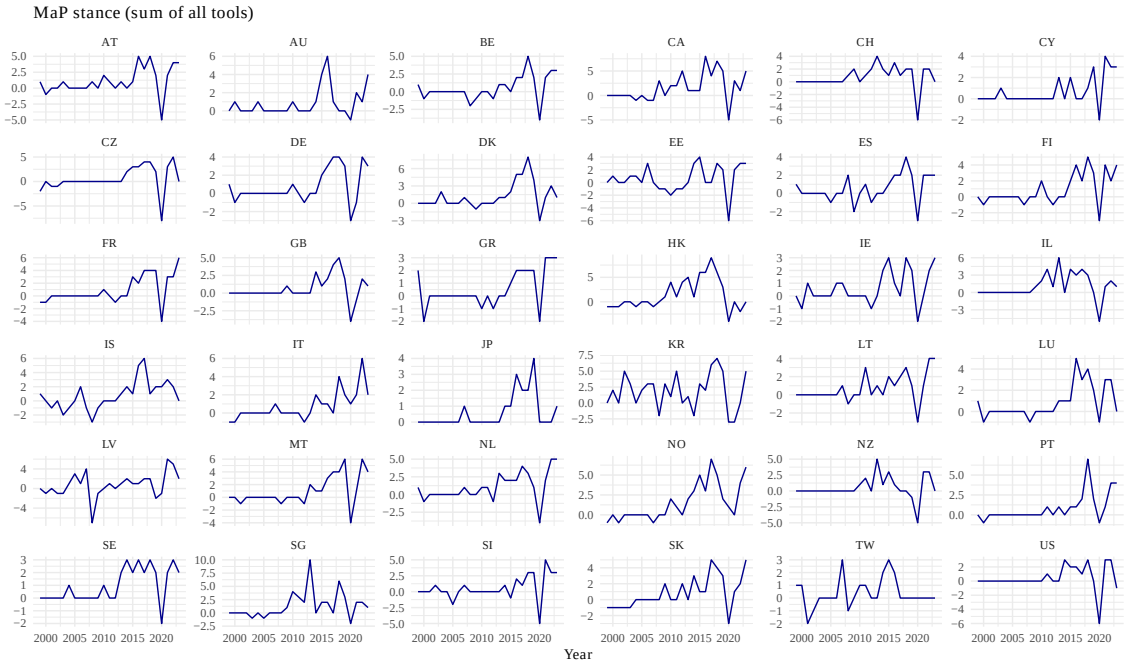
Notes: Data source: Standardized World Income Inequality Database (SWIID).

Figure 13: Disposable Gini Coefficient by Country - Developing Economies



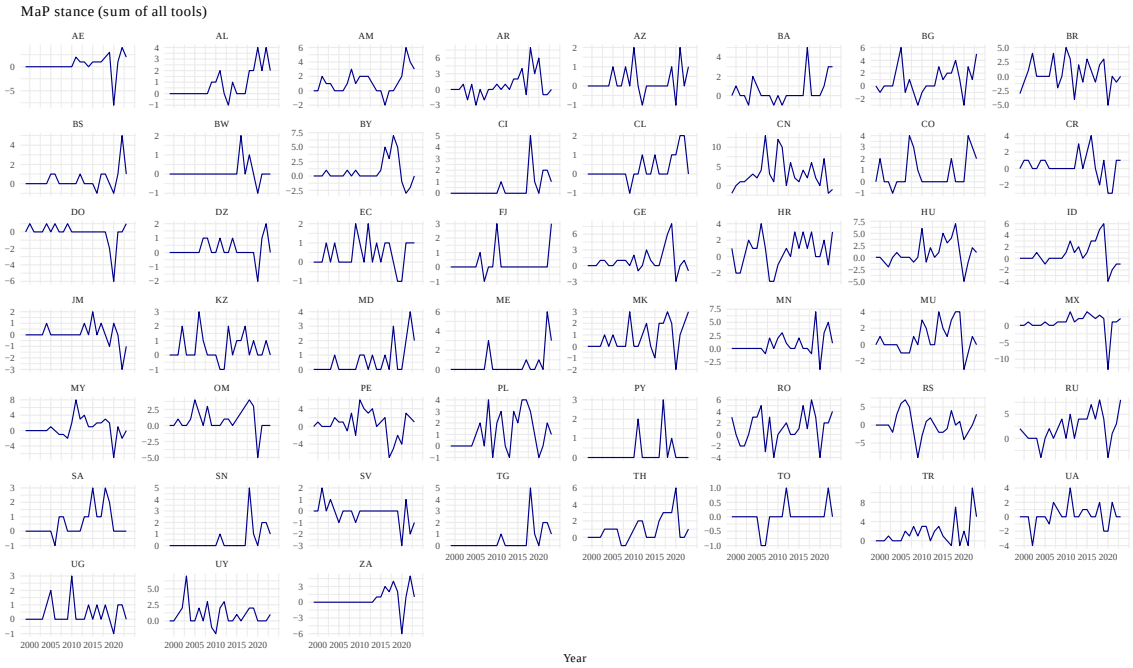
Notes: Data source: Standardized World Income Inequality Database (SWIID).

Figure 14: Macroprudential Policy Stance by Country - Advanced Economies



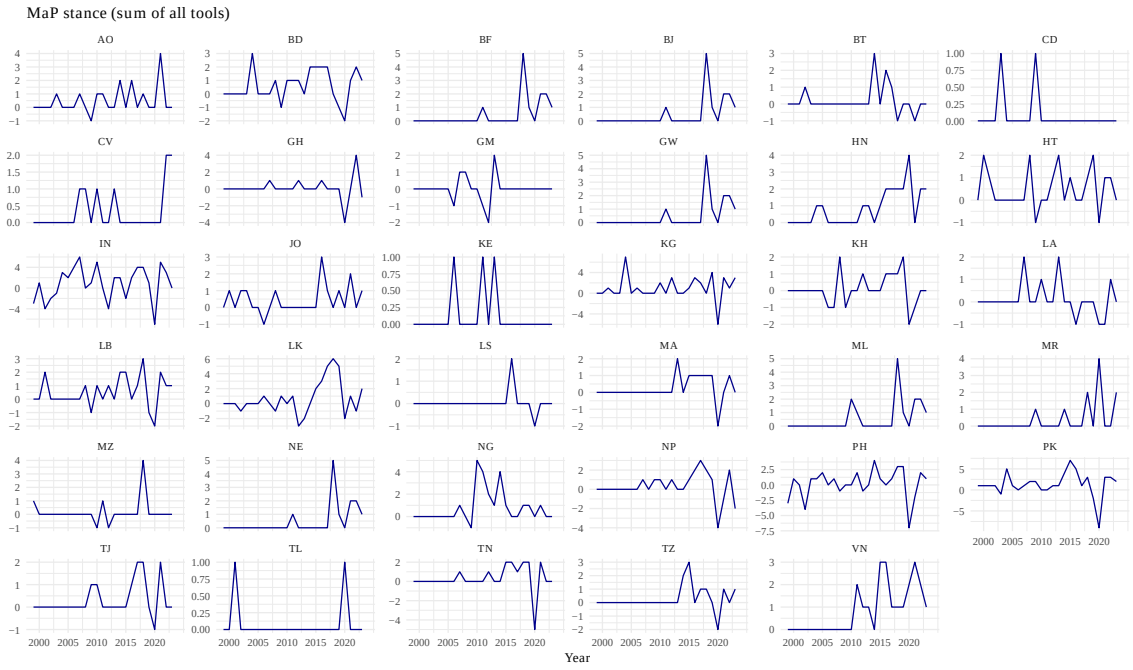
Notes: Data source: iMaPP–IMF (Alam et al., 2025). Each subplot displays the annual aggregate macroprudential policy index (sum of tightenings and loosenings across 17 instruments) for a single country.

Figure 15: Macroprudential Policy Stance by Country - Emerging Economies



Notes: Data source: iMaPP–IMF (Alam et al., 2025). Each subplot displays the annual aggregate macroprudential policy index (sum of tightenings and loosenings across 17 instruments) for a single country.

Figure 16: Macroprudential Policy Stance by Country - Developing Economies



Notes: Data source: iMaPP–IMF (Alam et al., 2025). Each subplot displays the annual aggregate macroprudential policy index (sum of tightenings and loosenings across 17 instruments) for a single country.

B Gini coefficient

This appendix derives a closed-form expression for the Gini coefficient implied by the model's two-agent structure, establishes an income-share representation that makes the transmission mechanism of macroprudential policy transparent, and establishes a sufficient condition under which a permanent tightening reduces steady-state inequality.

B.1 Steady-state Gini coefficient

Assumptions. We consider the deterministic steady state and normalize individual labor supplies to $h_h = h_e = 1$, so that aggregate labor supplies are $H_h = \mu$ and $H_e = 1 - \mu$. Budget constraints hold with equality and all asset positions are stationary. For the analytical steady-state characterization, we select the symmetric deposit allocation $d_e = d$ and normalize steady-state bond positions to zero, $b_h = b_e = 0$. We maintain the empirically relevant ordering $y_e > y_h$, which is satisfied at the calibrated steady state.

Profit terms. Using the profit-distribution definitions in Section 3.6, the banking-profit and fiscal components are distributed proportionally across the two groups, while entrepreneurs additionally receive profits from capital-goods production. At the deterministic steady state, $Q = 1$ and $I_t/I_{t-1} = 1$, so capital-producer profits vanish. It follows that the per-capita profit terms are equal across the two groups:

$$\pi_e = \pi. \tag{30}$$

Entrepreneur income. The entrepreneur's capital and labor earnings are, respectively,

$$\begin{aligned} w^k_{e,t} &= (R_{d,t-1} - 1)d_{e,t-1} + (R_{b,t-1} - 1)b_{e,t-1} + \pi_{e,t}, \\ w^l_{e,t} &= w_{e,t}h_e. \end{aligned}$$

The asset-income terms are written as net interest earnings because, at a stationary asset position, the principal carried from the previous period is rolled over into the current-period asset holding.

Applying the entrepreneur's budget constraint with $h_e = 1$, stationarity $d_{e,t} = d_{e,t-1} = d_e$, and the zero-bond normalization $b_e = 0$ gives

$$c_e = w_e + (R_d - 1)d_e + \pi_e.$$

Therefore the entrepreneur's per-capita steady-state income is

$$y_e \equiv w^k_e + w^l_e = w_e + (R_d - 1)d_e + \pi_e = c_e.$$

Household income. Similarly, household capital and labor earnings are

$$\begin{aligned} w^k_{h,t} &= (R_{d,t-1} - 1)d_{t-1} + (R_{b,t-1} - 1)b_{h,t-1} + \pi_t, \\ w^l_{h,t} &= w_{h,t}h_h. \end{aligned}$$

Using $h_h = 1$, stationarity $d_t = d_{t-1} = d$, and $b_h = 0$, the household budget constraint becomes

$$c_h = w_h + (R_d - 1)d + \pi,$$

so household per-capita steady-state income is

$$y_h \equiv w_h^k + w_h^l = w_h + (R_d - 1)d + \pi = c_h.$$

Using the steady-state property $d_e = d$ together with (30), the common deposit-income and profit components cancel from the income difference:

$$y_e - y_h = w_e - w_h. \tag{31}$$

Lemma B.1 (Closed-Form Gini Coefficient)**.** *Let*

$$\omega \equiv \frac{y_e}{y_h} = \frac{c_e}{c_h} > 1$$

denote the ratio of entrepreneur to household per-capita income at the steady state. Then

$$G = \frac{\mu(1-\mu)(\omega - 1)}{\mu + (1-\mu)\omega}. \tag{32}$$

Equivalently, defining the entrepreneur income share

$$\Lambda_e \equiv \frac{(1-\mu)y_e}{\mu y_h + (1-\mu)y_e} = \frac{(1-\mu)c_e}{\mu c_h + (1-\mu)c_e},$$

the Gini satisfies

$$G = \Lambda_e - (1-\mu). \tag{33}$$

Moreover, using (31) *and the steady-state resource constraint*

$$\mu c_h + (1-\mu)c_e = Y - \delta K,$$

the Gini can be written as

$$\boxed{G(K) = \frac{\mu(1-\mu)(w_e - w_h)}{Y - \delta K}}. \tag{34}$$

Proof. With population shares μ and $1-\mu$, per-capita incomes $y_h < y_e$, and mean income $\mu y_h + (1-\mu)y_e$, the two-group Gini coefficient is

$$G = \frac{\mu(1-\mu)(y_e - y_h)}{\mu y_h + (1-\mu)y_e}.$$

Substituting $y_e = \omega y_h$ yields (32). Alternatively, using the definition of Λ_e,

$$G = \frac{\mu(1-\mu)(y_e - y_h)}{\mu y_h + (1-\mu)y_e} = \Lambda_e - (1-\mu),$$

which establishes (33). Finally, under the steady-state portfolio selection $d_e = d$ and $b_h = b_e = 0$, equation (30) implies (31). Combining this with

$$\mu y_h + (1-\mu)y_e = \mu c_h + (1-\mu)c_e = Y - \delta K$$

gives (34). □

Equation (34) provides the starting point for the comparative-static analysis in Appendix B.2.

B.2 Comparative statics with respect to τ_k

We maintain the steady-state assumptions and portfolio selection introduced in Appendix B.1. In addition, the banking incentive constraint is assumed to bind, $\sigma > \rho$, and we restrict attention to an economically admissible steady state with $C = Y - \delta K > 0$ and $w_e > w_h$.

Proposition B.1 (Local inequality reduction under an explicit condition)**.** *Define the steady-state quantities*

$$a \equiv \frac{rK}{Y}, \qquad b \equiv \frac{rK}{rK + w_e H_e}, \qquad P \equiv \frac{w_e}{w_h}, \qquad x \equiv \frac{\delta K}{Y}. \tag{35}$$

A permanent increase in τ_k reduces the steady-state Gini coefficient locally if

$$\frac{a}{\sigma} + \frac{P}{P-1}\left(\frac{1}{\rho} - \frac{1}{\sigma}\right) b > \frac{a-x}{1-x}. \tag{36}$$

That is, if the elasticity of the wage gap with respect to capital exceeds the elasticity of aggregate disposable income. This condition holds at the baseline calibration. The result is local and conditional. It does not assert that $dG/d\tau_k < 0$ for every admissible parameter vector or tax rate.

Proof. From Lemma B.1 and the steady-state income accounting established in Appendix B.1, the steady-state Gini is given by (34). We therefore characterize how a permanent change in τ_k affects the capital stock, the wage gap, and aggregate disposable income.

Step 1: A tightening reduces the steady-state capital stock. At a deterministic steady state, both agents' stochastic discount factors equal β, and hence so does the bank's population-

weighted discount factor. Define

$$n \equiv \frac{N}{K}, \qquad R_d = \beta^{-1}, \qquad s \equiv (1-\tau_k)R_k - R_d,$$

and let total deposits be $\widetilde{D} \equiv D + D_e$. The bank balance sheet, net-worth dynamics, and binding incentive constraint imply

$$\widetilde{D} = (1-n)K, \qquad (1-\theta R_d)n = \theta s + \delta_T, \qquad j = \frac{\kappa}{n}. \tag{37}$$

Writing $\bar{\gamma}$ for the steady-state value of the multiplier γ_t on the incentive constraint, the two banking optimality conditions can be written as[16]

$$\beta(1-\theta+\theta j)s = \bar{\gamma}\kappa, \tag{38}$$

$$j(1-\bar{\gamma}) = \beta(1-\theta+\theta j)R_d. \tag{39}$$

Eliminating $\bar{\gamma}$ and substituting $j = \kappa/n$ yields

$$\kappa = \beta\left[(1-\theta)n + \theta\kappa\right]\left(R_d + \frac{s}{n}\right).$$

Using (37),

$$R_d + \frac{s}{n} = \frac{1-\delta_T/n}{\theta},$$

so n solves

$$\beta(1-\theta)n^2 + \left[\theta\kappa(\beta-1) - \beta(1-\theta)\delta_T\right]n - \beta\theta\kappa\delta_T = 0. \tag{40}$$

The leading coefficient of (40) is positive and the constant term is negative, so the quadratic has exactly one positive root. Its coefficients do not depend on τ_k; therefore, on the admissible branch, n is constant across tax-indexed steady states. Equation (37) then fixes s, and hence the required after-tax gross return on capital,

$$R \equiv (1-\tau_k)R_k = R_d + s > 0.$$

Since $R_k = r(K) + 1 - \delta$,

$$r(K) = \frac{R}{1-\tau_k} - 1 + \delta. \tag{41}$$

[16]Unlike the other variables this multiplier is barred to denote its steady-state value given γ already denotes the CRRA risk aversion parameter.

With fixed labor and technology, strict concavity of the nested-CES production function implies $r_K < 0$, where here and below a subscript K on r, Y, or V denotes the partial derivative with respect to K at the steady state, holding labor inputs and technology fixed (e.g., $r_K \equiv \partial r/\partial K$). Differentiating (41) therefore gives

$$\frac{dK}{d\tau_k} = \frac{R}{(1-\tau_k)^2 r_K} < 0. \tag{42}$$

Thus a permanent tightening lowers the steady-state capital stock. Notice that the marginal product r rises as capital falls.

Step 2: Compare the wage-gap and aggregate-income elasticities. Because $Y_K = r$, and using the definitions in (35), the output elasticity with respect to capital is

$$\frac{d\ln Y}{d\ln K} = a.$$

For the inner CES composite, Euler's theorem gives

$$\frac{d\ln V}{d\ln K} = \frac{KV_K}{V} = \frac{rK}{rK + w_e H_e} = b.$$

The wage equations then imply

$$\frac{d\ln w_h}{d\ln K} = \frac{a}{\sigma}, \tag{43}$$

$$\frac{d\ln w_e}{d\ln K} = \frac{a}{\sigma} + \left(\frac{1}{\rho} - \frac{1}{\sigma}\right) b. \tag{44}$$

Since $w_e > w_h$, the elasticity of the wage gap is

$$E_{\text{gap}} \equiv \frac{d\ln(w_e - w_h)}{d\ln K} = \frac{a}{\sigma} + \frac{P}{P-1}\left(\frac{1}{\rho} - \frac{1}{\sigma}\right) b. \tag{45}$$

The elasticity of aggregate disposable income is

$$E_C \equiv \frac{d\ln(Y - \delta K)}{d\ln K} = \frac{K(r-\delta)}{Y - \delta K} = \frac{a-x}{1-x}. \tag{46}$$

Differentiating (34) and using (45) and (46) therefore yields

$$\frac{dG}{d\tau_k} = \frac{G}{K}\left[E_{\text{gap}} - E_C\right]\frac{dK}{d\tau_k} = \frac{G}{K}\left[\frac{a}{\sigma} + \frac{P}{P-1}\left(\frac{1}{\rho} - \frac{1}{\sigma}\right) b - \frac{a-x}{1-x}\right]\frac{dK}{d\tau_k}. \tag{47}$$

Under condition (36), the term in square brackets in (47) is positive. Since $dK/d\tau_k < 0$ from (42), it follows that $dG/d\tau_k < 0$. □

The intuition behind the condition in (36) is that capital accumulation widens the wage gap proportionally more than it raises aggregate disposable income, which is expected given the complementarity between capital and high-skill labor. Consequently, a macroprudential policy tightening, which lowers K, compresses the gap faster than income, and the Gini falls.

Baseline verification. At the baseline calibration $\beta = 0.99$, $\theta = 0.972$, $\kappa = 0.384678$, $\delta_T = 0.002115$, $\delta = 0.025$, $\mu = 0.55$, $\alpha_u = 0.604820$, $\alpha_k = 0.227596$, $\sigma = 2.2$, $\rho = 1.26$, and $\tau_k = 0$. The reduced steady-state system gives

Quantity	Value	Quantity	Value
K	28.7175	a	0.3300
Y	3.2721	b	0.4720
r	0.0376012	P	1.5000
G	0.0867084	x	0.2194
E_{gap}	0.6302	E_C	0.1417

Thus $E_{\text{gap}} - E_C \simeq 0.4885 > 0$, so the sufficient condition in (36) is comfortably satisfied at the baseline. The implied local derivative is $dG/d\tau_k \simeq -2.3586$. Central finite differences using tax increments 10^{-4}, 10^{-5}, and 10^{-6} give -2.358621, -2.358596, and -2.358596, respectively, confirming the analytical derivative.

C Additional information on the model

C.1 Parameters

Table 3: Baseline Parameter Values

Parameter	Value	Description	Target / Source
Preferences and technology			
β	0.99	Discount factor	Standard quarterly
γ	2	Risk aversion	Standard
μ	0.55	Household population share	Li et al. (2023)
δ	0.025	Capital depreciation rate	Standard (10% annual)
ζ	3.456	Investment adjustment cost	Standard range
η_h, η_e	0.0025	Bond portfolio cost	Standard
Production — calibrated to match $w_e/w_h = 1.5$ *and* $rK/Y = 0.33$			
σ	2.2	Outer CES elasticity (unskilled vs. bundle)	Duffy et al. (2004)
ρ	1.26	Inner CES elasticity (capital vs. skilled)	Duffy et al. (2004)
α_u	0.6048	Low-skill labor weight	Skill premium $w_e/w_h = 1.5$
α_k	0.2276	Capital weight in skilled-capital nest	Capital income share $rK/Y = 0.33$
Banking — calibrated to match $K/N = 4$ *and annual credit spread* $= 100$ *bps*			
θ	0.9720	Banker survival probability	Gertler and Karadi (2011) Table 1
κ	0.3847	Bank diversion parameter	Implied by θ, K/N, and spread
δ_T	0.00212	Startup transfer to newborn bankers	Implied by K/N and spread
Shock processes			
$\rho_A,\ \sigma_A$	0.85, 0.007	TFP persistence and std. dev.	Smets and Wouters (2007)
$\rho_\xi,\ \sigma_\xi$	0.85, 0.005	Capital quality persistence and std. dev.	Gertler and Karadi (2011)
Policy rules			
ρ_τ	0.65	Policy rule inertia	
$\psi_\tau = \psi_K$	0.75	Feedback from primary target or capital	
ψ_G	0.25	Feedback from Gini secondary target	
σ_τ	0.001	Policy shock std. dev.	

Notes: Bond targets $\bar{b}_e = \bar{b}_h = 0$. All parameters held fixed across the policy specifications in Section 5. Production elasticities σ and ρ imply capital-skill complementarity ($\sigma > \rho$) in the sense of Krusell et al. (2000).

D Decomposition of the Gini responses

This appendix describes how we decompose the response of the model-implied Gini coefficient into the contributions of the different income sources of households and entrepreneurs. This is an accounting exercise rather than a structural decomposition: it does not attempt to identify the causal effects of each income source, but rather to describe how the movements in each source map into the Gini along the impulse responses of Section 5. The decomposition is complementary to the steady-state characterization in Appendix B. The exercise takes advantage of the fact that the components of the Gini, the income gap and mean income, are additive in the income sources.

D.1 Accounting decomposition of the Gini by income source

Let bars denote steady-state values. Following Section 3.7, the Gini coefficient is given by

$$G_t = \mu(1-\mu)\,\frac{g_t}{m_t}, \qquad g_t \equiv y_{e,t} - y_{h,t}, \qquad m_t \equiv (1-\mu)\,y_{e,t} + \mu\,y_{h,t}, \tag{48}$$

where g_t is the income gap between entrepreneurs and households and m_t is mean per-capita income. The income of each group is in turn the sum of its components, $y_{j,t} = \sum_k x^j_{k,t}$ for $j \in \{e, h\}$: labor income, bank payouts, tax rebates, deposit and bond income, and, for entrepreneurs only, the profits from capital-goods firms. Following Section 3.6.3, bank payouts and the rebated revenue are distributed equally in per-capita terms across the two groups.

Taking logs, the Gini becomes additive in the gap and mean income,

$$\ln G_t = \ln[\mu(1-\mu)] + \ln g_t - \ln m_t,$$

and a first-order log-linearization around the steady state yields

$$\frac{G_t - \bar{G}}{\bar{G}} \approx \frac{g_t - \bar{g}}{\bar{g}} - \frac{m_t - \bar{m}}{\bar{m}}.$$

Multiplying through by $\bar{G}$ expresses the response in absolute deviations of the Gini, the units used in Section 5:

$$G_t - \bar{G} \approx \bar{G}\left[\frac{g_t - \bar{g}}{\bar{g}} - \frac{m_t - \bar{m}}{\bar{m}}\right]. \tag{49}$$

Since both the gap and mean income are linear in the income components, their deviations can be split exactly into $g_t - \bar{g} = \sum_k \left(\Delta x^e_{k,t} - \Delta x^h_{k,t}\right)$ and $m_t - \bar{m} = \sum_k \left[(1-\mu)\Delta x^e_{k,t} + \mu\,\Delta x^h_{k,t}\right]$, with

$\Delta x^j_{k,t} \equiv x^j_{k,t} - \bar{x}^j_k$. Substituting into (49), the response of the Gini is the sum of the contributions of each component,

$$G_t - \bar{G} \approx \sum_k C_{k,t}, \qquad C_{k,t} = \bar{G}\,\frac{\Delta x^e_{k,t} - \Delta x^h_{k,t}}{\bar{g}} - \bar{G}\,\frac{(1-\mu)\,\Delta x^e_{k,t} + \mu\,\Delta x^h_{k,t}}{\bar{m}}. \tag{50}$$

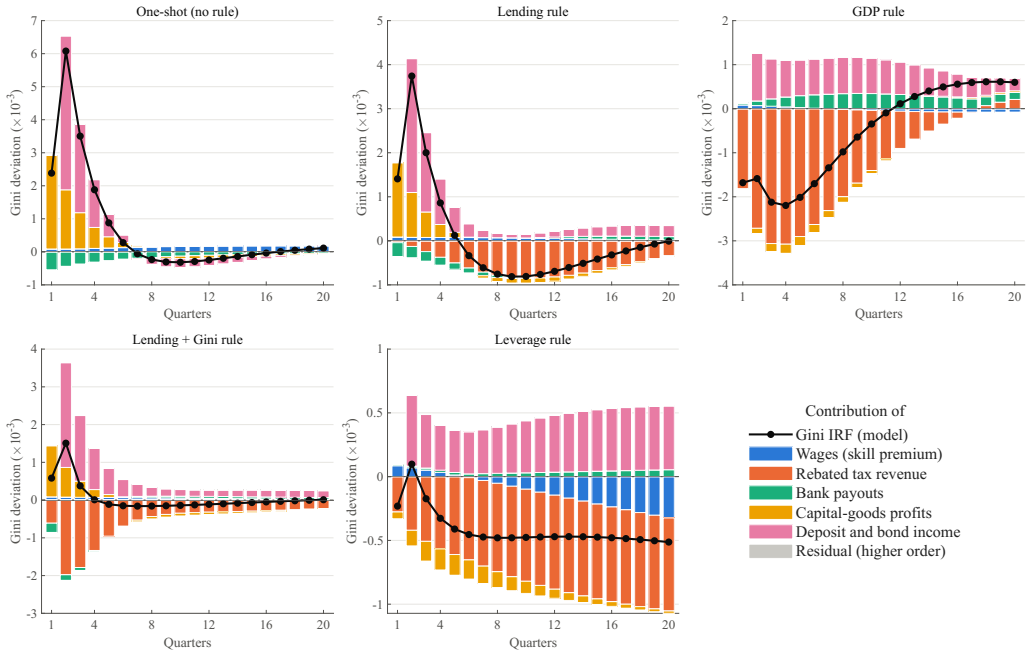
Equation (50) makes clear that each income source can affect inequality through two channels: by widening or narrowing the gap between the two groups, and by raising or lowering the mean income against which that gap is measured. Components that are equal in per-capita terms, such as bank payouts and the rebated tax revenue, operate only through the second channel, so that an increase in them lowers the Gini. Similarly, a proportional increase in both wages raises the gap and mean income in the same proportion, which implies that the labor-income component matters for the Gini only insofar as the skill premium changes.

We evaluate the deviations $\Delta x^j_{k,t}$ with the impulse responses of all model variables, setting lagged variables at their steady-state values in the period before the shock. Notice that the only approximation in this procedure is the log-linearization of the Gini in the gap and mean income, as the split of g_t and m_t into components is exact. The difference between the sum of the contributions and the model response is therefore a (small) higher-order residual.

D.2 Decomposition of the Gini responses by shock and policy rule

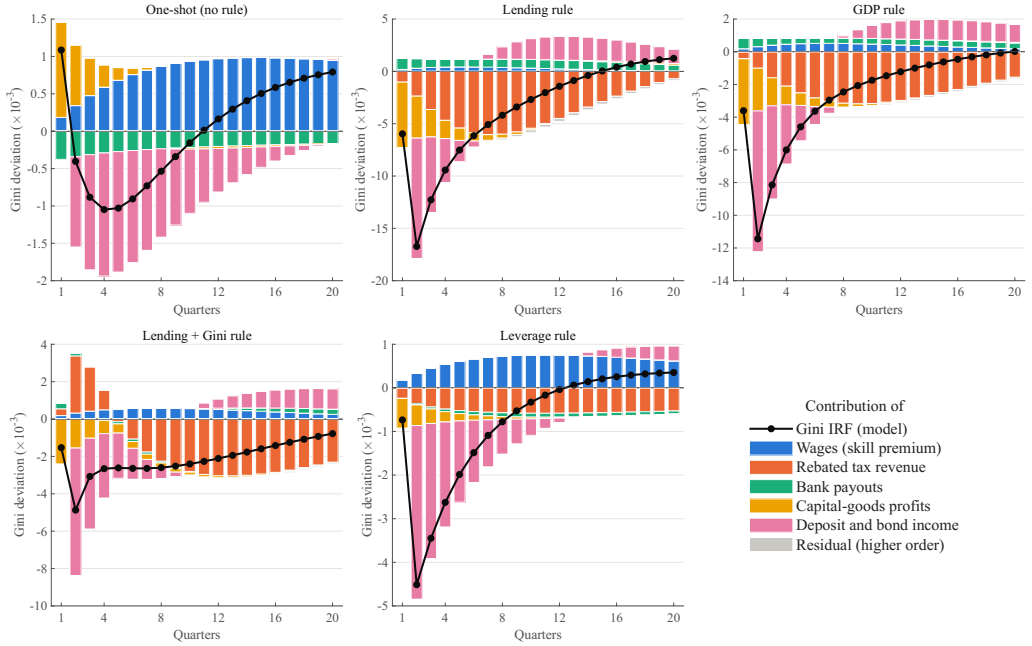
Figures 17–19 report the decomposition of the IRF in Section 5 for each policy specification and shock. In each panel, the black line with dots is the impulse response of the Gini, and the stacked bars are the contributions $C_{k,t}$ of the income components. The gray segment is the higher-order residual, so that the bars add up to the impulse response.

Figure 17: Decomposition of the Gini Response to a TFP Shock (ε_A)



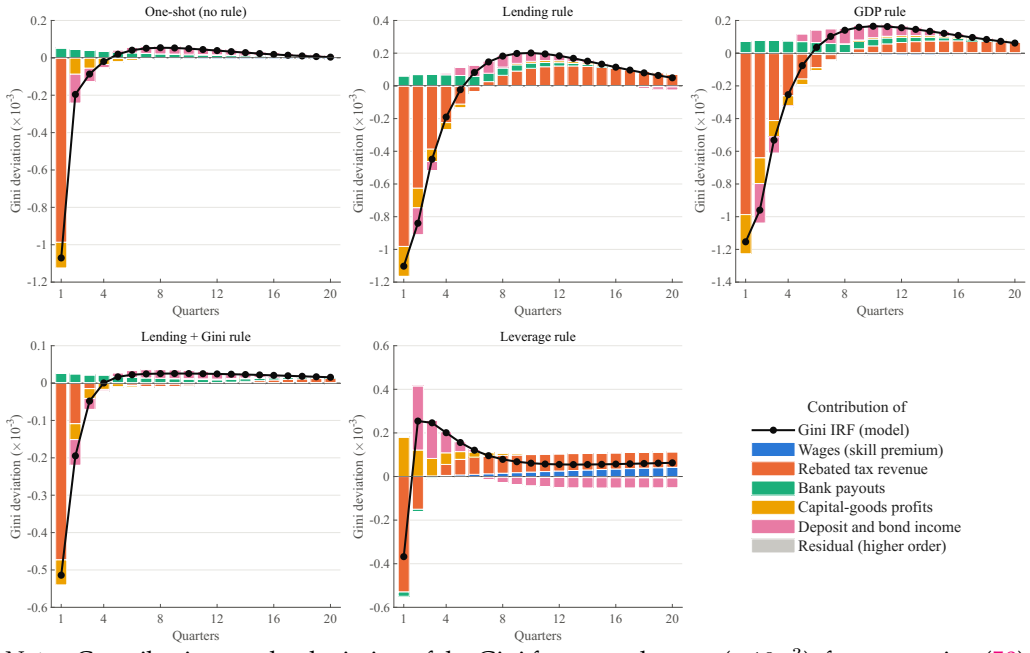
Notes: Contributions to the deviation of the Gini from steady state ($\times 10^{-3}$), from equation (50). Black line: Gini impulse response. Gray: higher-order residual. One-standard-deviation shock ($\sigma_A = 0.007$).

Figure 18: Decomposition of the Gini Response to a Capital Quality Shock (ε_ξ)



Notes: Contributions to the deviation of the Gini from steady state ($\times 10^{-3}$), from equation (50). Black line: Gini impulse response. Gray: higher-order residual. One-standard-deviation shock ($\sigma_\xi = 0.005$).

Figure 19: Decomposition of the Gini Response to a Policy Shock (ε_τ)



Notes: Contributions to the deviation of the Gini from steady state ($\times 10^{-3}$), from equation (50). Black line: Gini impulse response. Gray: higher-order residual. One-standard-deviation shock ($\sigma_\tau = 0.001$).